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Deciphering Deception:
How Different Rhetoric of AI Language Impacts
Users' Sense of Truth in LLMs

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Deciphering Deception: How Different Rhetoric of AI Language Impacts Users' Sense of Truth in LLMs

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TABLE OF CONTENTS

LIST OF TABLES.....	v
LIST OF FIGURES.....	vii
ACKNOWLEDGMENT.....	ix
ABSTRACT.....	x
CHAPTER 1. INTRODUCTION.....	1
CHAPTER 2. RELATED WORK.....	6
2.1. Deception and Credibility.....	6
2.1.1. Definition of Misinformation, Disinformation, and Deception.....	6
2.1.2. Credibility Assessment Theories and Heuristics.....	8
2.1.3. Landscape of Deception Detection.....	14
2.2. Deception Detection per Communication Channels.....	17
2.2.1. Deception Detection in Face-to-Face Communication...	17
2.2.2. Deception Detection in Computer-Mediated Communication.....	22
2.2.3. Deception Detection in AI	25
2.3. Rhetoric and LLM.....	32
2.3.1. Linguistic Studies in LLM.....	32

2.3.2. Rhetoric.....	34
2.3.3. Persuasive Rhetoric in ChatGPT.....	40
2.3.4. Hypotheses.....	42
CHAPTER 3. MATERIALS	51
3.1. Research Prototype.....	51
3.2. Questionnaire in AI Prototype.....	52
3.3. Responses in AI Prototype.....	53
CHAPTER 4. STUDY DESIGN.....	59
4.1. Participants	59
4.2. Tasks and Procedures	60
4.3. Analysis Methods	63
4.3.1. Quantitative Analysis	63
4.3.2. Qualitative Analysis	64
CHAPTER 5. RESULTS.....	66
5.1. Quantitative Analysis in Block 1.....	66
5.1.1. Differences in Truth Discernment by Type.....	66
5.1.2. Sensitivity and Specificity Analysis of Varying Types....	69
5.1.3. Compared perception for each AI System	71
5.1.4. Impact of Control Variables on Truth Discernment Across AI Systems.....	73

5.2. Quantitative Analysis in Block 2.....	76
5.2.1. Differences in Truth Discernment by Type.....	76
5.2.2. Sensitivity and Specificity Analysis of Varying Rhetoric Types	79
5.2.3. Impact of Control Variables on Truth Discernment Across AI Systems	80
5.3. Qualitative Analysis	83
5.3.1. Truth and False Cues in Block 1.....	83
5.3.2. Expectations for the AI System	86
5.3.3. Truth and False Cues in Block 2.....	88
5.3.4. Expectancy for AI System	89
CHAPTER 6. DISCUSSION.....	92
6.1. Findings.....	92
6.1.1. Hypotheses Evaluation.....	92
6.1.2. Human Perception of AI.....	96
6.1.3. Contextual Influences on Trust and Perception of AI...	100
6.1.4. Evaluation Process	100
6.1.5. Contribution.....	103
6.2. Limitations	104
6.3. Suggestion for Future Work.....	106
CHAPTER 7. CONCLUSION.....	111

REFERENCES.....	116
국문요약.....	137

LIST OF TABLES

Table 1. Misinformation and Disinformation.....	8
Table 2. Credibility Assessment Theories.....	13
Table 3. Ethos Element Added to the Experiment.....	46
Table 4. Logos Element Added to the Experiment	47
Table 5. Results of Repeated Measures ANOVA for Block 1.....	68
Table 6. Participants' Truth Discernment Ability by Type and Truthfulness of Question per AI System in Block 1	69
Table 7. Sensitivity and specificity per each type in block 1.....	70
Table 8. ANOVA Results for Truthfulness, Logicality, and Transparency.....	72
Table 9. ANOVA Results for Human-likeness and Understanding Thoughts and Emotions.....	73
Table 10. Results of the Linear Mixed-Effects Model for Google in Block 1	76
Table 11. Results of the Linear Mixed-Effects Model for ChatGPT in Block 1.....	76
Table 12. Results of the Linear Mixed-Effects Model for NR in Block 1...	76
Table 13. Results of Repeated Measures ANOVA for Block 2.....	78
Table 14. Participants' Truth Discernment Ability by Type and Truthfulness of Question per AI System in Block 2.....	79

Table 15. Sensitivity and Specificity per Each Type in Block 2.....	80
Table 16. Results of the Linear Mixed-Effects Model for Default ChatGPT in Block 2.....	82
Table 17. Results of the Linear Mixed-Effects Model for Self-correction in Block 2.....	82
Table 18. Results of the Linear Mixed-Effects Model for Counterargument in Block 2.....	82
Table 19. Results of the Linear Mixed-Effects Model for Footnote in Block 2.....	83
Table 20. Truth Cues Used by the Participants to Determine the Truth of the AI response.....	84
Table 21. False Cues Used by the Participants to Determine the Falsehood of the AI Response.....	85

LIST OF FIGURES

Figure 1. Study Framework from Rhetorical and Heuristic Perspectives.....	4
Figure 2. Rhetorical Example in ChatGPT.....	41
Figure 3. Examples of Questionnaire.....	53
Figure 4. Examples of Answers for Block 1.....	55
Figure 5. Examples of Answers for Block 2.....	58
Figure 6. Three-Stage Study Process: Two Experiments Followed by an Interview.....	59
Figure 7. Experimental Setups Showcasing the AI Prototype and Online Survey Form Presented Adjacently.....	60
Figure 8. AI Prototype Identified by Participants' Email Address and Embedded Preset Questions.....	62
Figure 9. Comparison of Truth Discernment among AI Systems: ChatGPT Shows the Highest Score and Significant Difference Between True and False Answers.....	68
Figure 10. Comparison of Truthfulness, Logicity, and Transparency among Three Systems in Block 1.....	72
Figure 11. Comparison of Human-likeness and Understanding Thoughts and Emotions among Three Systems in Block 1.....	73
Figure 12. Coefficients of a Linear Mixed-Effects Model for Each AI System in Block 1.....	75

Figure 13. Comparison of Truth Discernment among AI Systems: Footnote Shows the Highest Score and Significant Difference Between True and False Answers.....	78
Figure 14. Coefficients of a Linear Mixed-Effects Model for Each AI System in Block 2	81
Figure 15. Adaptive Evaluation Process for AI-Generated Language.....	103

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Abstract

Deciphering Deception: How Different Rhetoric of AI Language Impacts Users' Sense of Truth in LLMs

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With the rise of generative AI like ChatGPT, users are increasingly exposed to AI-generated language, presenting potential deception and communication risks. This study delved into the rhetorical aspect of AI-generated language influencing users' truth discernment. The research conducted two quantitative user studies with experiments comparing users' ability to discern truth across three different rhetorical conditions: ChatGPT, Google, and No Rhetoric (Block 1), along with AI models incorporating four different persuasive rhetorical elements: GPT default, Self-correction, Counterargument, and Footnote (Block 2). Additionally, qualitative interviews were used to identify the reasoning behind their truth judgments regarding various AI systems. Results showed that outputs with fewer rhetorical elements challenged users to distinguish truth from false, while those with more rhetoric often misled users into false truths. Users' AI expectations also influenced their truth judgments. Responses meeting users' expectations were deemed truthful, while those not meeting expectations were seen as less credible. Statements giving a human-like impression were often perceived as false. People expected the AI system to

represent itself as an AI with technical, precise responses rather than evoke a human image. Among varied rhetorical specifics, footnotes in responses triggered a notably high truth bias, even when the answer was false. Meanwhile, including a counterargument enabled users to leverage their prior knowledge effectively, leading to more accurate judgments. In Human AI communication, people often rely on persuasive rhetoric, leading to intuitive judgments based on linguistic heuristics rather than careful deliberation when discerning truth from falsehood. Participants tend to perceive AI as objective and superior, favoring its responses over their knowledge. However, encountering errors in AI responses leads to negative perceptions towards AI, indicating a form of distrust and bias when AI fails to meet expectations. For improved transparency and truth discernment in AI-generated language, AI designs should judiciously employ rhetorical elements and set guidelines to minimize potentially deceptive responses, including proactive measures to warn about cognitive vulnerabilities and ensure source accuracy. Additionally, individuals need to develop digital literacy skills to evaluate information critically, recognize signals of accurate truth discernment, and raise awareness of AI's potential to produce falsehoods, addressing human cognitive biases effectively.

Keywords: AI-generated language; Human-AI communication; Deception detection; Rhetoric; Human AI perception; Heuristics

Chapter 1. Introduction

ChatGPT has garnered significant attention for its adeptness in providing comprehensive and contextually accurate answers to a diverse range of user queries, fostering optimism about the bright future text-based generative AI might offer (Guo et al., 2023). However, concerns have arisen due to the occasional inaccuracies, incorrectness, and hallucinations in ChatGPT's responses (Borji, 2023), leading to fears of user deception, fake news, and manipulation. Inherently, models like GPT operate within the limitations of their design as language models. Instead of constructing answers based on verified facts, they operate probabilistically, selecting words based on their likelihood to follow previous words (Bender et al., 2021), and as such, inherently do not distinguish between truth and falsehood. Recognizing these critiques, OpenAI issues a disclaimer to users, warning that "the system may occasionally generate incorrect or misleading information and produce offensive or biased content." However, such warnings do not indicate when the system might provide false information, nor do they offer guidelines on how to handle such occurrences. Without a means to overcome these issues, users might overlook the benefits and potentials of ChatGPT, using the service with skepticism and in a limited manner.

Historically, the domain of human communication has been meticulously scrutinized, with particular emphasis on the art and science of deception detection (DePaulo et al., 2003; Luke, 2019; Vrij, 2000). It is commonly known that people occasionally struggle to discern truth in face-to-face conversations (Vrij et al., 2010).

Even when considering both verbal and non-verbal cues, the ability to accurately detect deception barely exceeds 50% (Rubin et al., 2015). This conundrum is exacerbated within the Human-AI communicative paradigm, wherein such non-verbal cues are conspicuously absent. Existing literature underscores the intricacy of discerning truthfulness from fabricated statements generated by ChatGPT (Danry et al., 2022). ChatGPT's persuasive strategies may complicate the discernment of truth in its responses. As we anticipate more frequent interaction with advanced AI systems engaging in contextual and intelligent conversations, it will be necessary to understand our perceptions and evaluate methods for detecting deception. Generative AI systems, like ChatGPT, heavily rely on language and are mediated by rhetoric (Zdenek, 2003), making it crucial to understand the language's underlying heuristics. In these systems, language is not just a means but also the primary value, form, and cue. However, there has been a lack of rigorous discussions on leveraging rhetoric as an effective tool to diminish deceptive tendencies and promote transparent interactions.

This observation propelled the research and led to the postulation of a series of hypotheses. Central to the inquiry is understanding the user's navigational strategy in the AI conversational domain, particularly in an environment bereft of standard non-verbal signals. How do individuals distinguish truthfulness in AI-generated language, which cannot provide non-verbal and behavioral cues, unlike human conversations (RQ1)? How do various types of persuasive rhetorical elements influence truth discernment (RQ2)? How do people exhibit cognitive judgment tendencies towards language-based generative

AI responses (RQ3)? What individual characteristics influence people's discernment of truth and falsehood in language-based generative AI (RQ4)?

To explore these, the study pursued several stages of inquiry. Initially, an investigation delved into the definitions and theories regarding deception and credibility of information, theories concerning truth discernment in deception detection, and assessing their relevance in face-to-face communication, computer-based communication, and interactions with AI. Subsequently, the study investigated people's perceptions of AI and linguistic heuristics and research needs for text-based AI, concentrating on language usage patterns within ChatGPT.

Then, a user study with experiments and interviews was conducted to scrutinize the impact of rhetorical elements on users' truth discernment during interactions with ChatGPT. The study designed a ChatGPT simulator working on AI models containing three levels of rhetorical elements: ChatGPT, Google, No Rhetoric (Block 1), and AI models containing four varying rhetorical elements: GPT default, Self-correction, Counterargument, and Footnote (Block 2) for the experiments. The semi-structured interview captured participants' perceptions and insights into conversational AI systems, including their preferences and expectations. The investigation also explored variations in truth discernment abilities among models, participants' subjective criteria, and decision-making processes. Furthermore, an examination was conducted into how users' perceptions, expectations of AI, linguistic bias, and heuristics collectively play a crucial role in truth discernment.

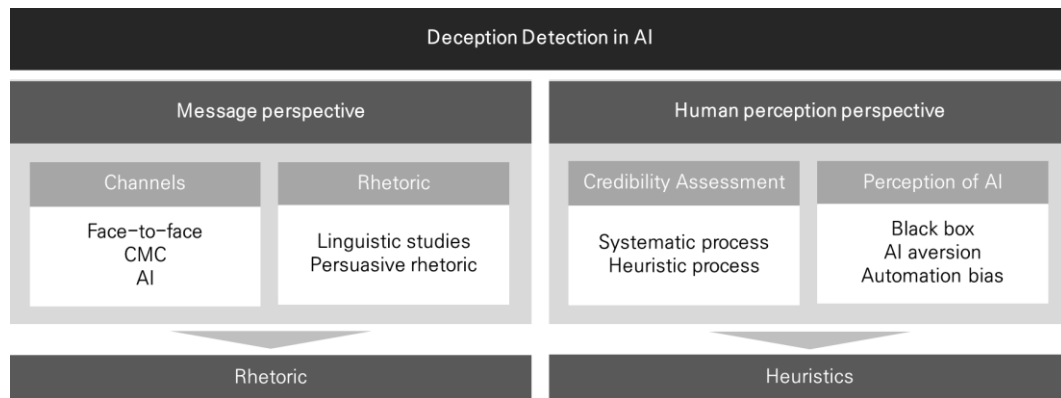


Figure. 1. Study Framework from Rhetorical and Heuristic Perspectives

The results of the study indicated the following:

- Rhetoric in AI-generated language affects truth discernment.
- Limited rhetoric impedes truth differentiation, while excessive rhetoric can lead to truth bias.
- Users tend to make immediate judgments on AI-generated language using linguistic heuristics rather than conscious analytical decision-making.
- Users perceive AI responses as truthful, logical, and transparent.
- Users prefer AI to be technical rather than human-like in truth discernment tasks.
- The alignment between users' expectations of AI and their actual experiences is critical in determining their trust in AI outputs.

This study elucidated what brings about differences in the ability to discern truth in Human-AI communication, the rhetorical elements, and other factors at play. It clarified what subjective cues for truth and falsehood users bring into conversations with AI. It also examined the subjective bias of expectations towards AI and its role in complicating truth discernment. The study explored the theories applicable to the deception detection process in human-AI interaction and suggested directions for future research to enhance users' ability to discern truth in conversations with AI systems.

The research findings propose three significant contributions to academia and industry. First, it advocates for research from a rhetorical perspective to address issues in text-based generative AI to promote transparent communication, highlight the impact of rhetorical techniques on user judgment, and emphasize the need for linguistic research, guidelines, and regulations. Second, it provides insights into cognitive heuristics and biases in human assessment of AI language to design text-based AI design in the industry. Finally, it underscores the importance of educating individuals about accurate comprehension of AI and digital literacy, as well as the necessity for social and institutional rules and regulations.

Chapter 2. Related Work

2.1. Deception and Credibility

2.1.1. Definition of Misinformation, Disinformation, and Deception

In exploring how individuals discern the truthfulness of specific information, this study begins with a foundational inquiry into the cognitive definitions of information and deception. Information is defined from a social constructionist perspective as a “communicative construct produced within a social context”. This definition, as articulated by Tuominen and Savolainen (1997), is handy and comprehensive for discussions on misinformation and disinformation, emphasizing the role of communicative acts in constructing information (Karlova & Fisher, 2013). Throughout the process of communication, information can be categorized as disinformation or misinformation based on the range/level of its accuracy, completeness, and currency, as well as the presence or absence of intent to deceive on the part of the information provider (Karlova & Fisher, 2013; Pennycook & Rand, 2021).

Disinformation refers to information that is false or inaccurate, specifically created with the intention to deceive others (Pennycook & Rand, 2021). Fallis (2009) suggests that while disinformation is typically inaccurate, it may not always be so and could also exhibit contextually misleading traits. The intent behind disinformation can vary, encompassing motives ranging from social benevolence to personal, adversarial reasons (Karlova & Fisher, 2013).

Misinformation refers to false, inaccurate, or misleading information that, although not intended to deceive, ultimately conveys untruthful information (Pennycook & Rand, 2021). Karlova and Lee (2011) further elaborate that misinformation is characterized by inaccuracies, uncertainties (which could involve presenting multiple possibilities or choices), vagueness (lack of clarity), or ambiguity (being open to multiple interpretations) (Karlova & Fisher, 2013).

The informativeness of disinformation and misinformation can depend on how the recipient perceives the information. Buckland (1991) emphasized that "being informative is situational", highlighting that the informativeness of information changes based on the recipient's situation, knowledge, and context. This perspective underscores that what is considered misinformation in one scenario may not be viewed as such in another due to differing interpretations and understandings, thereby illustrating the complex nature of identifying and defining misinformation (Karlova & Fisher, 2013).

Deception is a message that is knowingly transmitted by a sender with the intention of fostering a false belief or conclusion in the receiver (Buller & Burgoon, 1996). Deception can be defined on the premise that there are both a sender and a receiver of information. Deception can also be conceptualized as occurring when an identifiable actor's intention to mislead leads to attitudinal or behavioral changes that align with this intention (Chadwick & Stanyer, 2022). This notion is anchored in the understanding that the essence of deception is not merely in the intent to mislead but in the successful achievement of that intent. This perspective facilitates a more comprehensive approach to

studying disinformation, misinformation, and misperceptions, proposing a holistic framework for understanding these phenomena (Chadwick & Stanyer, 2022).

According to the definitions provided, when communication agents are human, the distinction between misinformation and disinformation can be made based on the presence or absence of intent. However, categorizing information from sources whose authors are unknown, such as web or social media content, is more complex. This study aims to examine the truthfulness of AI-generated content, such as that produced by Large Language Models (LLMs) like ChatGPT. Instances of hallucination generated by LLMs are currently not considered to stem from an intent to deceive but instead are viewed as structural flaws of the LLM systems. Therefore, this paper treats such instances under the broader term of misinformation. When users of LLMs perceive the inaccuracies in AI-generated language content as factual, despite its inaccuracies, this falls under the category of misinformation from an informational perspective, and the message can be defined as deceptive from the recipient's standpoint.

	Misinformation	Disinformation
Accuracy	Yes or No	Yes or No
Complete	Yes or No	Yes or No
Current	Yes or No	Yes or No
Deceptive Intention	No	Yes

Table. 1. Misinformation and Disinformation

2.1.2. *Credibility Assessment Theories and Heuristics*

As the internet flooded the late 1990s with new information, the risk of encountering diverse forms of misinformation and disinformation in daily life increased.

The importance of accurately discerning the truthfulness of intentionally falsified or ambiguously contextualized information, determining the level of trust, and understanding the process by which individuals make such judgments has become increasingly critical. This problem created an urgent need to verify information within interactive environments, and the concept of credibility received significant attention (Hilligoss & Rieh, 2008). Credibility has been defined in various ways, including terms such as believability, trust, reliability, accuracy, fairness, and objectivity (Self, 1996). It has also been characterized in relation to the traits of persuasive sources, the structure and content of the message, and perceptions of the media (Hilligoss & Rieh, 2008; Metzger et al., 2003).

Theories to understand the stages of credibility assessment are diverse. Fogg's (2003) Prominence-Interpretation theory posits that users undergo two distinct processes: noticing something (prominence) and then making judgments about what they have noticed (interpretation). Fogg's theory highlights that the interpretation and judgment process is shaped by a variety of influences, including the cultural background, previous experiences, and heuristic thinking of the user, their proficiency in the subject matter of the site, and the surrounding circumstances such as the expectations, and situational norms of the user (Hilligoss & Rieh, 2008). Wathen and Burkell's (2002) judgment model explains the process in three stages: the first stage assesses the credibility of a medium based on surface characteristics such as appearance/presentation, usability/interface design, and information structure. The second stage involves evaluating the source and

the message, and the third stage deals with how the content interacts with the user's cognitive state. Sundar's (2007) model highlights the importance of cognitive heuristics related to credibility assessment in digital media, presenting four "affordances" that can detect credibility: Modality (M), Agency (A), Interactivity (I), and Navigability (N). Sundar argued that these affordances could amplify or diminish the content's effect on credibility, acting as moderators through various psychological means (Hilligoss & Rieh, 2008). Hilligoss and Rieh (2008) identified three key elements in credibility assessment: construct, heuristics, and interaction, to develop an integrated framework for assessing credibility. Constructs are broad concepts shaping judgment; heuristics are general rules applied across situations; and interactions focus on judgments based on specific cues, simplifying the framework's complexity.

One significant theoretical perspective in credibility assessment is the Elaboration Likelihood Model (ELM), which addresses the factors and effects involved in the cognitive processes that can occur when attempting to change a person's attitude through communication, as well as the intensity of the resulting judgments (Petty et al., 1986). The Elaboration Likelihood Model (ELM) suggests that individuals follow different processing routes when evaluating messages, influenced by their motivation and ability. When motivation and the ability for systematic thinking are high, individuals are likely to follow the "central route" of persuasion. However, when motivation is low or cognitive abilities are limited, people tend to follow the "peripheral route" of persuasion (Petty et al., 1986). Individuals' cognitive engagement with messages varies depending on the

degree of motivation for message processing and the level of ability to process the message. Motivation can be influenced by personal relevance to the topic, a preference for thoughtful consideration, and a sense of responsibility for the outcomes of message processing. The ability to process messages is contingent upon factors such as available time, level of knowledge, environmental conditions, and the frequency of message repetition (Rucker & Petty, 2006; Wagner & Petty, 2011). Due to a lack of motivation or capability for systematic processing, individuals often rely on heuristic processing, making judgments based on peripheral cues from the message or its source (Petty et al., 1986; Sundar, 2007).

Another critical theory influencing credibility judgment for information is the Dual Process Theory, which posits that humans have two cognitive processes during information processing. These two processes include Type 1 processing, which is fast, unconscious, and judgment based on heuristic knowledge, and Type 2 processing, which is slow, conscious, and decision-making based on analytical judgments (Evans & Stanovich, 2013). Type 1 processing is autonomous and occurs as a reflexive reaction when a stimulus arises (Stanovich, 2004, 2009a, 2011). Type 2 processing involves cognitive higher-order reasoning processes that require effort, allowing us to engage in "hypothetical thinking" and "cognitive decoupling" without confusing the real world with imagined situations (Stanovich, 2009b, 2011). Type 2 thinking provides a unique form of cognition that uniquely defines the human species, allowing for the definition of humans as a species characterized by unique cognitive forms (Evans, 2010b; Evans &

Over, 1996; Stanovich, 1999, 2004, 2011). The recent perspective of Default Interventionism suggests that Type 1 generates the default response until Type 2 intervenes (Evans & Stanovich, 2013). People tend to be generally lazy in using cognitive resources; however, when the outcome of decision-making is risky or highly important, they resort to Type 2 processing (Evans & Stanovich, 2013). When confronted with new problems, people often generate intuitive responses quickly and effortlessly. However, these responses may be inappropriate when individuals lack relevant experience. Therefore, in some instances during the reasoning and decision-making process, it is necessary to inventory initial intuitions and effectively substitute Type 2, namely deliberative reasoning (Evans & Stanovich, 2013).

Among the theories related to credibility assessment, particularly the Elaboration Likelihood Model (ELM) and Dual Process Theory have been employed by credibility researchers (Eastin et al., 2006; Sundar, 2007; Wathen & Burkell, 2002) to understand the process and elements of credibility assessment (Hilligoss & Rieh, 2008). According to the Elaboration Likelihood Model (ELM) and Dual Process Theory, consumers of digital news are known to follow a quick and intuitive heuristic mode rather than engaging in deep reflection amidst information saturation (Lang, 2000). During such judgments, decisions are primarily based on cues in the message or the way the message is presented (Bryanov & Vziatysheva, 2021) rather than undergoing conscious deliberation for every piece of information. This entails adopting efficient judgment through simple rules of thumb (heuristics) and utilizing limited cognitive resources. (Sundar, 2008). Heuristics

are experiential strategies useful for decision-making and problem-solving, especially in situations requiring quick and efficient decisions (Gigerenzer, 2008). Even when individuals may not fully comprehend the content of a message, they can rely on various heuristics to judge the credibility of the message. The factors influencing these heuristics encompass social influence through popularity and celebrity endorsement, the relevance of the topic, and the format in which the information is presented (Bryanov & Vziatyshcheva, 2021; Lewandowsky et al., 2019).

Authors	Theories	Description
Fogg (2003)	Prominence-Interpretation model	1) noticing something (prominence) 2) making judgments (interpretation)
Wathen & Burkell (2002)	(Three stages model)	1) medium credibility based on appearance and interface 2) evaluating source and message 3) content interaction with user cognition
Sundar (2007)	MAIN model	Modality, Agency, Interactivity, and Navigability
Hilligoss & Rieh (2008)	Credibility assessment framework	Construct, Heuristics, and Interaction
Petty et al. (1986)	Elaboration Likelihood Model	1) Central route: systematic processing 2) Peripheral route: heuristic processing
Stanovich (2004)	Dual Process Theory	Type 1) fast, unconscious, heuristic knowledge Type 2) slow, conscious, analytical judgments

Table. 2. Credibility Assessment Theories

As revealed by the aforementioned theories, in the process of assessing credibility, humans have been shown to exhibit cognitive biases that do not adhere to principles of logic, probabilistic reasoning, and validity. Psychology attributes these biases to inappropriate use of heuristics due to limited data and lack of specialized knowledge (Korteling et al., 2018). Heuristics are problem-solving methods based on personal experience, involving examining a limited number of signals or choices to make

decisions. Heuristics reduce the workload of information retrieval and storage and simplify decision-making by reducing the amount of integrated information required for decisions (Dale, S., 2015). However, heuristics can also lead to errors and biased judgments (Dale, S., 2015).

In neuroscience, when investigating decision-making and truth discernment, human cognitive biases and heuristics are explained as physiological traits, representing unique mechanisms within the brain's neural networks. These biases stem from fundamental characteristics of neural information processing aimed at optimizing functions such as perception-motor tasks, pattern recognition, and stability maintenance, which represent aspects of human heritage that have survived throughout human evolutionary history. Examples include a tendency to seek relationships and patterns (Association), prioritize the present and the relevant (Compatibility), judgments through retained information (Retainment), and concentrate on available salient information (Focus) (Korteling et al., 2018). While these biases can be highly efficient in executing perception-motor tasks, they can also lead to various distortions in cognitive information processing (Korteling et al., 2018; Korteling & Toet, 2022). Due to these cognitive characteristics of humans, judgments related to truth discernment also entail the potential for errors in judgment due to the use of heuristics rather than relying on logic, probabilistic reasoning, and validity (Johnson Jr, 2014).

2.1.3. Landscape of Deception Detection

The rise of misinformation and disinformation underscores the critical need to accurately discern truth from deception and understand individuals' judgment processes. Recent research in the field of deception detection has focused significantly on fake news, which has become a pressing social issue due to the ease with which false online digital information can be created and spread, posing a significant threat. Deception detection in AI-generated language shares commonalities with fake news, as both deal with misinformation/disinformation of varying degrees of accuracy and completeness, ultimately requiring correction to prevent the obfuscation of users' judgment. Therefore, this paper can draw upon various studies in the field of fake news when discussing the detection of truth in digital information. In the realm of fake news, deception detection is viewed from three perspectives: the message itself, the characteristics of the receiver of the message, and the environment in which the message is disseminated (Bryanov & Vziatysheva, 2021).

Firstly, regarding the message itself, studies have explored how users evaluate the truthfulness of a message based on its characteristics, such as consistency, presentation style and format, subject matter, source attribution, alignment with pre-existing beliefs, and social endorsement (Bryanov & Vziatysheva, 2021). It has been observed that users are more inclined to perceive messages as truthful when the topics align with their pre-existing beliefs or when the content has social endorsement (Kim et al., 2019; Luo et al., 2022). However, contradictory findings have also been reported regarding the same factors (Clayton et al., 2019; Schaewitz et al., 2020), and there are challenges in

generalizing characteristics due to the manipulation of experimental variables and environments (Bryanov & Vziatysheva, 2021).

Secondly, research has also focused on individual characteristics and human reasoning processes involved in discerning the truthfulness of information. This area of study often revolves around the Dual Process Theory, indicating that motivated Type 2 reasoning, which is engaged when there is a high need for cognition, can lead to more accurate truth discernment (Bago et al., 2020; Martel et al., 2020; Schaewitz et al., 2020). It's known that there are significant individual differences among those who are deceived by fake news, with tendencies toward higher bullshit receptivity, a greater inclination to overclaim knowledge, and lower scores on the Cognitive Reflection Test (CRT), which measures analytical thinking (Pennycook & Rand, 2020). These individuals also exhibit reflexive openness and rely on heuristics such as familiarity. Studies suggest that education or training can help people shift from reflexive to reflective thinking to counteract deception.

Finally, critical digital literacy and interventions have been identified as environmental elements influencing deception detection (Bryanov & Vziatysheva, 2021). Generally, providing guidelines that prompt users to reflect on the accuracy of news has improved the ability to discern truthfulness (Guess et al., 2020). However, contrasting studies suggest that generic warning messages could lead to decreased trust in legitimate news (Van Duyn & Collier, 2019), indicating the need for caution and further research in applying such guidelines (Bryanov & Vziatysheva, 2021).

Among these three aspects, this study aims to explore how the linguistic characteristics inherent in AI-generated language influence individuals' ability to discern truth, focusing on the rhetorical aspects of the message itself. The study does not overlook individual characteristics and environmental factors but intends to measure users' perceptual changes based on the message format as a critical focus of the investigation. The goal is to identify the elements influencing the evaluation of truthfulness and review extensive research in communication to understand how truthfulness is perceived within the message. By investigating traditional methods of truth discernment and reviewing studies on deception detection in face-to-face conversations and computer-mediated communication (CMC), the research also explores the current state of deception detection in AI and the importance of linguistic research in this context.

2.2. Deception Detection per Communication Channels

2.2.1. Deception Detection in Face-to-Face Communication

Trust and honesty in conversations are fundamental prerequisites for communication (Dubberly & Pangaro, 2009; Grice, 1975). Thus, numerous research has been conducted regarding deception detection (DePaulo et al., 2003; Luke, 2019; Vrij, 2000). Conversations occur through interactions between a sender and a receiver, and differentiating deceptive conversations involves the receiver interpreting various cues from the sender (Buller et al., 1991). In human-to-human conversations, people use cues like nodding, changing postures, and stutters to judge the truthfulness of the conversation

(Buller et al., 1991). However, despite people's overconfidence in their ability to detect lies, their actual ability is only around 54% (Bond Jr & DePaulo, 2006; Levine, 2010; Vrij et al., 2019). This is because participants in a conversation tend to assume trust in others to facilitate the conversation, which impairs their ability to detect lies (Buller et al., 1991).

In the realm of deception detection in communication, scholars have extensively explored various factors including non-verbal/behavioral cues, verbal cues, and the motivation of the deceiver. Historically, there was a prevalent belief that lies could be discerned through behavioral cues such as heightened vocal tension, increased pitch, pauses, speech errors, and rigid posture (Ekman, 2009). However, recent meta-analyses have indicated that non-verbal cues are unreliable for predicting deception (Masip, 2017). In practical scenarios, relying on contextual details such as tangible evidence, third-party insights, and inconsistencies in the deceiver's statements compared to known facts proves more effective (Park et al., 2002).

Consequently, the focus shifted away from behavioral cues, and endeavors commenced to discern genuine behavioral distinctions between truth-tellers and deceivers via interview methodologies (Masip, 2017). One approach is the 'verifiability approach,' which found that liars provided significantly fewer details when asked for verifiable details compared to truth-tellers (Nahari et al., 2014). Another is the 'cognitive load approach,' which suggests that lying increases cognitive load, leading to inconsistent or insufficient answers (Vrij & Fisher, 2016). Studies explaining deception detection

through verbal cues include the Reality Memory and CBCA approaches. The Reality Memory approach suggests that actual experiences yield more contextual and sensory details (Masip, 2017). According to RM theory, differences in verbal hedge frequency, self-references, and word count significantly distinguished conditions from utterly false to truthful confessions (Zhou et al., 2004).

Similarly, the CBCA method, employed in child abuse detection, finds that firsthand experiences tend to meet a set of criteria related to incident details, thoroughness, and context (Raskin & Esplin, 1991). The liar's motivation also affects the receiver's ability to detect deception. Lies told by highly motivated people are more likely to be detected than those by less motivated individuals (Vrij, 2000).

The ability to detect deception can also be influenced by the fact that humans have inherent cognitive constraints and predispositions (Gilbert, 1991). Such innate tendencies impact decision-making and the deception detection process (Hauch et al., 2015). People rely on cognitive heuristics to make judgments about the veracity of statements (Levine & McCornack, 2001), tend to focus excessively on nonverbal cues rather than verbal ones (Reinhard et al., 2011), and are biased towards either truth or lie, regardless of the actual veracity of a statement (Levine et al., 1999).

Given the inherent biases and limitations in human judgment, it is essential to delve deeper into the theoretical frameworks underlying deception detection. One theory that aims to explain the cognitive process of detecting deception is the Interpersonal deception theory (IDT). It explains deception is a dynamic process involving interactions

between the deceiver and the receiver, shaped by personal goals and the evolving dynamics of communication (Buller & Burgoon, 1996). According to IDT, senders of deceptive messages often experience anxiety about the potential detection of falsehoods in their messages. Receivers, on the other hand, attempt to detect the truthfulness and validity of information, harboring suspicions about the presence of deception (Buller & Burgoon, 1996).

The theory identifies three elements in a deceptive message: the core message (primarily expressed linguistically), the adjunct message (involving verbal and non-verbal cues that often reveal the truthfulness of a specific message), and incidental behaviors (mostly non-verbal actions that inadvertently "leak" the deceiver's intentions) (Buller & Burgoon, 1996). Deceptive individuals often exhibit reticence by using fewer words and less information, and they tend to employ less clear, inconsistent sentence structures while conveying semantically indirect or irrelevant messages (Burgoon et al., 1996). They also distance themselves from the message, using fewer first-person pronouns or passive voice constructions. (Zhou et al., 2004).

Another theory is the Truth-Default Theory (TDT). According to TDT, people start with a passive presumption of honesty towards others. If they can't find active evidence of deception, they basically assume that the statement is true (Levine, 2014; Markowitz & Hancock, 2023). This theory delineates conversation into content, context, and demeanor and emphasizes the importance of 'coherence' (logical consistency of

content) and 'correspondence' (alignment of content with existing knowledge) as key factors in detecting deception (Levine, 2014)

Alternatively, the Adaptive Lie Detector (ALIED) theory posits that people don't inherently lean toward truth-bias; instead, they discern truth contextually. This approach suggests an adaptive judgment process relying on two distinct factors. If highly diagnostic individuating cues allow falsehood to be detected, then that evidence is given priority for deception detection. If cues are insufficient for making a decision, context-general information plays a more decisive role in the final decision (Bond Jr et al., 2013). The adaptive process can explain why, in environments where falsehoods are more commonly stated (such as criminal confessions), a lie bias rather than a truth bias may emerge. According to this theory, to enhance detection accuracy, the diagnosticity of individuating cues should be heightened, either by using effective questioning methods (Levine, 2014) or by scrutinizing what underlying motives or benefits the declarant might have (Bond Jr et al., 2013).

Additionally, Language Expectancy Theory (LET) describes the cognitive processes involved in persuasion during communication and what circumstances are individuals predisposed to favor truthfulness or deception. It defines communication behavior as evolving appropriately based on a society's psychological and cultural norms. The use of language that does not conform to these norms affects the behavior and attitudes of the message receiver (Burgoon et al., 2002). The theory explains why the linguistic formats of a message affect the results of persuasion and describes people as

beings affected by unconventional language styles that differ from their expectations (Burgoon et al., 2002). According to this theory, people determine the communicator's credibility and the statement's truthfulness based on whether the language used matches their expectations.

2.2.2. Deception Detection in Computer-Mediated Communication

Historically, research on deception detection centered around face-to-face interactions. However, with the rise of digital communication methods, including emails, social networks, intelligent replies, and auto-complete features, the focus has shifted towards deciphering deceit within computer-mediated communication (CMC). In text-based CMC, the verbal message becomes almost the sole and primary medium for communication, thereby making detecting truth even more challenging (Hancock et al., 2010). Also, this indicates a significant reliance on language for discerning truth from falsehood, and language exerts a considerable impact on individual perceptions and emotional judgments.

Vrij (2000) posits that the absence of nonverbal and vocal cues in CMC could result in diminished capacities for detecting deception compared to face-to-face interactions. Empirical research supports this notion, demonstrating that individuals often perceive statements made by liars in text-based interactions as more truthful than those made by truth-tellers (Burgoon et al., 2003). In another research, individuals with higher motivations to deceive are often more successful in text-based interactions, which stands

in contrast to face-to-face communication, where higher deceptive motivation can lead to easier detection (Hancock et al., 2010).

One of the significant characteristics of CMC is the controllability afforded by its features of editability and increased composition time (Hancock et al., 2010). Crafting deceptive information inherently demands greater cognitive effort than telling the truth (Zuckerman et al., 1981). Hence, the CMC environment, which allows for extended response time periods, provides an advantageous setting for those with deceptive intentions (Burgoon & Floyd, 2000). The option to meticulously edit messages before transmission also affords deceivers a significant control advantage (Hancock et al., 2010).

To overcome the challenges of truth discernment and limited deception detection capability in computer-mediated communication (CMC), various technical methods for truth discernment have been developed, including voice stress analyzers, EEGs, and polygraphs (Hauch et al., 2015). Attempts have also been made to use Linguistics-Based Cues (LBC) for deception detection. LBC attempts to discern deception using linguistic information from one or more word-based units, relying on main linguistic cues derived from existing theoretical backgrounds such as CBCA (Criteria-based content analysis), and RM (Reality monitoring). These indicators include passive voice usage, self-referencing, negativity, broad terms, uncertainty, time-related details, spatial details, and emotional expressions (Zhou et al., 2002). Experimental results from various linguistic constructs such as quantity, diversity, complexity, specificity, expressivity, informality, affect, uncertainty, and non-immediacy revealed that deceptive messages tend to be

longer, less formal, ambiguous, simple, monotonous, and more expressive (Zhou et al, 2004). These results diverged from theories of face-to-face communication; the observation that deceivers tend to use more language contradicts (IDT) prediction that deceivers generally choose less quantity, while the finding of more emotional information among deceivers contradicts RM research claiming that deceivers are less expressive and emotionless. The absence of evidence supporting spatial-temporal information contradicts predictions from CBCA and RM (Zhou et al., 2004).

Many other studies have been conducted to discern truth using linguistic-based cues, among which one study conducted a meta-analysis to investigate whether computers can detect linguistic cues associated with deception. The study revealed that deceivers express more negative emotions, use fewer sensory-perceptual words, and mention cognitive processes less frequently than those who say truths than those who say truths (Hauch et al., 2015). Additionally, research has demonstrated accuracy exceeding 80% in automated tests that classify criminal narratives, interrogations, and legal testimonies into truth and lies based on linguistic indicators (Bachenko et al., 2008). There is also research comparing the linguistic characteristics of satire, hoaxes, and propaganda in news media to identify language features of unreliable texts (Guo et al., 2022). These ongoing studies demonstrate the utility of linguistic cues in deception detection and the positive signals of automated deception detection.

However, overall studies based on linguistic cues have not reached a consistent conclusion on which linguistic indicators are consistently effective in deception detection,

primarily due to the context-sensitive nature of text-based cues (Bryanov & Vziatysheva, 2021; Rubin & Conroy, 2012). Many experiments have been conducted under strictly controlled conditions, such as specific contexts, topics, message formats, and lengths, making generalization of experimental results difficult when these experimental environments change (Rubin & Conroy, 2012). Changes in the characteristics and context of experiments, and the real-time nature of response generation, may lead to different deception avoidance strategies employed by users, potentially altering the characteristics of factors identified as Linguistics-Based Cues (Zhou et al., 2004). Methods for truth discrimination based on linguistic cues face the challenge of developing general automated deception methods or systems capable of adapting to various contexts. Therefore, there have been other efforts, such as exploring deception through systematic content analysis, considering content theme, centrality, realism, essence, and self-distancing, among other factors (Rubin & Conroy, 2012).

2.2.3. Deception Detection in AI

With the advancement of AI, we are now encountering various forms of AI services in our daily lives, such as virtual assistants, autocompletes, and smart replies. In this era, it has become crucial to discern the veracity of information provided by AI and ascertain its accuracy. However, deception detection in AI interactions, which involves more complex factors beyond computer-mediated communication, requires examination of how individuals perceive AI, as it significantly affects the dynamics of human-AI interaction and decision-making processes. Recent research has identified perceptions

toward AI including views on black box systems, AI aversion, and automation bias.

Generative AI produces responses through interactions among billions of parameters and user input values, making it extremely challenging to comprehend the processes and decision-making involved. AI demonstrates outstanding predictive and decision-making performance, yet the average user often lacks a clear understanding of the underlying mechanisms and processes through which AI generates such responses and may not always trust the results. This lack of knowledge about AI and direct/concrete interaction leads to reliance on immediate judgments and mental heuristics rather than deep thinking when evaluating AI (Fiske & Taylor, 2013; Just & Latzer, 2018). Due to the perception of AI as a black box and limited interaction with AI systems, perceptions of AI are subjective and constrained (Just & Latzer, 2018; Shin & Park, 2019). Consequently, people develop different heuristics based on context and individual differences when perceiving AI (Bellur & Sundar, 2014).

Research has shown that AI algorithms demonstrate greater prediction accuracy than humans. However, there is a tendency among people to trust human forecasters more than high-performing algorithms. Even though AI has demonstrated better performance, when aware that the algorithm has made a mistake, individuals exhibit algorithm aversion, refusing to use the system (Alvarado-Valencia & Barrero, 2014; Bucher, 2017; Dietvorst et al., 2015). Numerous studies indicate that people generally prefer human advice over AI advice, especially in contexts such as ethical decisions, medical choices, and pleasurable experiences, demonstrating a reluctance to rely on AI/algorithms (Böhm,

2023). Researchers attribute algorithm aversion to the perception that AI's flawless execution is undermined and its inability to adequately reflect contextual factors (Dietvorst et al., 2015; Grove & Meehl, 1996; Highhouse, 2008).

However, other studies suggest that algorithm aversion is not uniformly manifested and depends on the context in which the algorithm is used, with some cases even preferring AI decisions (Castelo et al., 2019; Jones-Jang & Park, 2023; Sundar & Kim, 2019;). In one study, when individuals were informed about the source of AI-generated advice, they slightly downgraded the author's capabilities and values, but it did not affect their willingness to follow and share AI advice on social issues (Böhm et al., 2023). Additionally, positive experiences with generative AI, such as ChatGPT, have reduced AI aversion (Böhm et al., 2023). In cases where content was known to be created by AI rather than human experts, users tended to perceive the quality more favorably, indicating human favoritism rather than AI aversion (Zhang & Gosline, 2023). Thus, the emergence of AI, such as ChatGPT, appears to mitigate AI aversion or alter its context.

Studies indicate that, even though AI possesses both human-like qualities (Nass et al., 1991) and machine-like characteristics (Jones-Jang & Park, 2023), people tend to map the characteristics of interaction with AI more towards machine-like features than human-like features (Jones-Jang & Park, 2023; Shin & Park, 2019; Sundar, 2020). Common perception people hold about AI is that it is pre-programmed, objective, and consistent (Jones-Jang & Park, 2023; Sundar & Kim, 2019). Moreover, various studies point to the presence of Automation bias toward AI, where individuals tend to

excessively overestimate the accuracy and consistency of AI due to its superior performance (Cummings, 2017; Jones-Jang & Park, 2023; Lee, 2018). Consequently, people often trust stories generated by AI or messages verified by AI to be more objective and unbiased compared to those created by humans (Jones-Jang & Park, 2023; Wojcieszak et al., 2021).

Individuals with these diverse perceptions of AI, engage in various cognitive and emotional processes to assess the truthfulness of AI-generated language. Deception detection in AI interactions, which is a level beyond computer-mediated communication, involves more complex factors. Firstly, state-of-the-art AIs like ChatGPT have an inherent problem of hallucination. Large Language Models like ChatGPT generate answers by setting high-priority words based on a massive amount of language data and integrating information. Instead of being based on knowledge or facts, they create sentences by inferring the next most likely word in a linguistic format. They sometimes fabricate facts and provide answers that are not true. This is known as the characteristics of 'hallucination' or 'stochastic parroting' (Bender et al., 2021) and there is still no clear solution to fix these errors. When using explanatory AI systems like ChatGPT interactively, the users show similar levels of trustworthiness even when providing false explanations compared to when providing true explanations (Danry et al., 2022). There is a pressing need for extensive research to understand how humans cognitively perceive and react to the falsehoods generated by AI systems.

Secondly, cognitive deception can arise due to the linguistic explanations created by AI systems based on Natural Language Processing. This form of deception is not about perceiving falsehood as truth but refers to a broader cognitive error. It stems from linguistic formats or perceiving and trusting the conversational partner as an entity that comprehends and resonates with human emotions and thoughts. When a linguistic explanation is provided to the user, it affects their perception and behavior, even if it is superficial (Langer et al., 1978).

For example, the chatbot Eliza, created by Joseph Weizenbaum in 1964, made the conversational partners believe that the chatbot has the intelligence to understand their thoughts and emotions, even though it lacked true AI capabilities (Russell & Norvig, 2010). Weizenbaum used a Rogerian argument (a discourse strategy developed by Carl Rogers designed not to win the argument but to enhance bilateral communication) to create Eliza's conversation patterns. It involved finding relevant keywords or patterns from the input language from the conversation partner and providing empathetic responses based on specific rules, making users feel that Eliza is an intelligent AI (Natale, 2019). Weizenbaum demonstrated that through the use of discourse, he created the illusion of intelligence and revealed that simple rhetorical language use can make users vulnerable to deception in interactions with computers (Weizenbaum, 1966).

Unlike Eliza, the latest AI, ChatGPT, is a functioning real AI that inherently possesses various deceptive characteristics. Recent research has highlighted these deceptive traits within ChatGPT. Noever and Ciolino (2022) conducted a series of

experiments using ChatGPT-3 and made people believe that they were talking to another human, not AI, showing the potential for chatbots to be used for deceptive purposes (Liu et al., 2023; Noever & Ciolino, 2022). Khalil and Er (2023) examined the perceived originality of various writings produced using GPT by verifying them with two plagiarism detection tools, Turnitin and iThenticate. They found that GPT produces output that is difficult to detect by software and revealed it to have a level of originality equivalent to humans (Khalil & Er, 2023).

Hence, there has been a growing recognition of the potential socio-economic impact of deceptive aspects of LLMs, leading to multidimensional research efforts aimed at studying and mitigating LLMs' misinformation and disinformation generation. Research indicates that misinformation generated by LLMs, even with the same semantics, can be more challenging to detect than human-created misinformation (Chen & Shu, 2023a; Pérez-Rosas et al., 2017). This suggests that LLMs possess a more deceptive style and have a greater potential for exploitation (Chen & Shu, 2023a). People can detect only 9.6% of misinformation created by hallucinations, and finding inaccuracies in data, such as dates, addresses, names, and citations, proves exceedingly difficult (Chen & Shu, 2023a). LLMs can generate various style of information through manipulating prompting more effectively, making it difficult to detect the maliciously generated LLM information using existing detectors.

Hence, ongoing efforts such as selecting training data and processes (Casper et al., 2023; Yu et al., 2020) and prompt filtering (Jain et al., 2023) are essential to mitigate

the misinformation generated by LLMs. Additionally, there is a continuous need for endeavors such as developing detectors (Chern et al., 2023; Jiang et al., 2023; Kaliyar et al., 2021; Zhou et al., 2023) to detect misinformation created by LLMs. Recently, there has been mention of the term 'TrustLLM,' focusing on trustworthiness principles, evaluation criteria, and related discussions regarding LLMs' trust, as noted by Sun et al. (2024). Projects such as Do-Not-Answer, HELM, and HaluEval are underway to design, develop metrics, test, and evaluate trustworthy models, aiming to create transparent and reliable LLMs (Liang et al., 2022; Li et al., 2023; Wang. Y. et al., 2023). Moreover, research on enhancing awareness of the limitations of LLMs in generating false information and educating users about the characteristics and identification methods of misinformation are executed (Chen & Shu, 2023b).

The call for national-level efforts to build trust in LLMs is growing louder; however, current regulations primarily focus on conventional AI models, with insufficient regulation specifically tailored for LLMs (Sun et al., 2024). Due to risks that may arise from privacy concerns during the training process of LLMs, demands have been made in Italy to suspend LLM services, and efforts are underway in the EU to draft digital service acts in 2023 (Sun et al., 2024). The National Institute of Standards and Technology (NIST) in the United States has published guidelines for managing risks associated with LLMs through an AI Risk Management Framework (Sun et al., 2024).

In the industry, efforts should be made to understand and address potential risks

associated with LLMs in all aspects of research and development, product production and distribution, and user experience management. These efforts include fair use of data in research and development, verification procedures to prevent biases, and mechanisms to filter AI-generated content and prevent the spread of harmful content. Furthermore, it is necessary to establish clear ethical principles regarding aspects such as data privacy, transparency, fairness, and societal impact. It is important to educate users on how AI systems operate, their functions, biases, and issues, and provide users with means of control such as privacy settings, transparency features, and content filtering to ensure data privacy.

2.3. Rhetoric and LLM

2.3.1. Linguistic Studies in LLM

This increasing presence of deceptive attributes in ChatGPT necessitates closer examination. In text-based AI systems like ChatGPT, language serves as the primary user interface and communication channel. Recent studies are actively investigating the linguistic attributes of ChatGPT, a system known for its evidence-backed and systematic responses, which represents a novel form of human-AI communication, thereby increasing the potential for deception and manipulation. As language and rhetoric mediate these generative AI systems (Brahnam, 2009; Zdenek, 2003), it is essential to analyze the heuristics underpinning our linguistic interactions.

Several studies have compared natural language and AI-generated language to identify parallels and contrasts. ChatGPT mimics human linguistic patterns across various psycholinguistic tasks (Cai et al., 2023). In analyzing essay creation tasks, AI content, characterized by a structured, scientific style, often surpasses human writing in quality (Herbold et al., 2023). In news text generation tasks, humans exhibit more varied vocabulary and more dispersed sentence length distribution compared to LLMs, which use more numbers, symbols, and pronouns (Muñoz-Ortiz et al., 2024). In political debate generation tasks, LLMs can generate original and new political claims and are expressed more readably and positively compared to human-generated text (Palmer & Spirling, 2023).

Moreover, studies on human responses to ChatGPT outputs reveal interesting biases. While humans can typically distinguish between human and AI texts with repeated exposure (Guo et al., 2023), specific topics such as first-person narratives about family and humanity blur this discernment, pointing to avenues for AI-led deception (Jakesch et al., 2023). While ChatGPT emulates human language to some extent, it exhibits unique characteristics distinct from human prose. Furthermore, various studies have examined linguistic aspects of LLMs, such as how manipulation of LLM prompting's readability, formality, and concreteness affects the reduction of hallucination (Rawte et al., 2023). Another study found that language patterns predicting individual characteristics can amplify or adjust their impact depending on the language context, the specific LLM used, and the adopted prompt (Sourati et al., 2024). In the field of natural

language processing, researchers conducted research on linguistic inference tasks of LLMs (Wang, B et al., 2023). Remarkably, very little studies have ventured into the realm of the rhetorical perspective for AI-generated language in deception detection. This research seeks to bridge that gap by examining the perception of AI language through the perspective of persuasive rhetoric.

2.3.2. Rhetoric

Rhetoric is the study of using language to persuade others. It has played a role in various fields since the classical era of Greece and Rome, including human history, culture, and social interactions (Herrick, 2020). Rhetoric encompasses extensive research on how language is used for persuasion, how messages are structured and delivered, and how we communicate and convey information in everyday interactions (Burke, 1969; Hartelius & Browning, 2008). Constructivism sees communication from the perspective of cognitive processes, suggesting that individuals vary in their ability to create and comprehend cognitively complex messages during communication. It posits that individuals who are cognitively complex may be more successful in communication due to their ability to use rhetorical design logic when crafting messages (Delia & O'Keefe, 1982). The utilization of rhetoric and construction of persuasive message in communication is considered an essential element for successful delivery and effective communication.

Aristotle, who introduced this concept of rhetoric, emphasized three pillars of persuasion in his book 'Ars Rhetorica': Ethos (speaker's credibility), Logos (reason and

logic), and Pathos (appeal to audience's emotion) (Bartlett, 2019). Aristotle's rhetoric has influenced much of modern persuasion research (Larson, 1989). With the development of mass communication in the 20th century, there arose a greater need for more detailed studies on rhetoric and persuasion (Demirdöğen, 2010). In the 1950s, the Yale study group conducted extensive experiments to understand the principles and processes of persuasion (Hovland et al., 1953). Using rhetorical variables as independent variables and examining the impact of each variable on changes in the attitude of message receivers as the dependent variable persuasion (Demirdöğen, 2010). They encompass three factors: (1) the source of persuasive communication, (2) the attributes of the message, and (3) the traits of the recipient and the message's context, corresponding to Aristotle's Ethos, Pathos, and logos as such these theories are referred to as the New Rhetoric (Demirdöğen, 2010; Smith, 1981). The following sections will explore the definitions of these three elements in both historical and contemporary contexts.

Ethos signifies the listener's trust in a speaker (Giffin, 1967). Aristotle conceptualized Ethos with three dimensions: intelligence, character, and goodwill (McCroskey & Young, 1981). Aristotle views Ethos as "the charisma and credibility of the speaker," considering non-verbal aspects such as the speaker's appearance and reputation as factors influencing Ethos (Demirdöğen, 2010). In the Yale study, Ethos factors affecting persuasion included the communicator's credibility, expertise, likability, and similarity to the audience (Demirdöğen, 2010). Subsequent definitions by other researchers have expanded Ethos to encompass authority, character, and expertise (Green

Jr, 2004; Higgins & Walker, 2019) and is described as revolving around persuasion through credibility and trustworthiness (Hartelius & Browning, 2008; Koszowy et al., 2022). To establish a robust Ethos, various elements such as the speaker's expertise, similitude, ingratiation, self-criticism, inclination to succeed, and consistency should resonate throughout the discourse (Higgins & Walker, 2019). From a linguistic perspective, the significance of examining personal pronouns and other linguistic indicators to reveal the Ethos conveyed by speakers is highlighted (Mshvenieradze, 2013; Ting, 2018).

Ethos is also known by various terms such as prestige, reputation, and authority (Applbaum & Anatol, 1972; Giffin, 1967; McCroskey, 1966), but the most used term is "source credibility," which refers to positive characteristics of the communicator that influence the receiver's acceptance of a message. Source credibility, defined by Hovland et al. (1953), encompasses expertise and trustworthiness. Expertise refers to the extent to which a communicator is perceived as a reliable source of valid assertions, while trustworthiness is the level of confidence in the communicator's intent to convey the most valid assertions (Ohanian, 1990).

When the communicator is perceived as highly trustworthy, an opinionated message tends to be more effective in inducing attitude change than a non-opinionated communication (McGinnies & Ward, 1980). Expertise, encompassing authoritativeness, competence, expertness, and qualification (McCroskey, 1966; Berlo et al., 1969), has been identified as crucial in influencing receivers' attitude change based on the perceived

expertise of the information source (Crano, 1970; Crisci & Kassinove, 1973; Ohanian, 1990). Various scales have been developed to measure source credibility, and factors for assessing trustworthiness include dependability, honesty, reliability, sincerity, and trustworthiness, while expertise dimensions include expert, experienced, knowledgeable, qualified, and skilled (Ohanian, 1990).

In this digital age, one concept that has been discussed concerning Ethos is cognitive authority. Since the emergence of the web, it has become challenging to assess the quality and authority of digital information, leading to active discussions on the concept of cognitive authority, which emphasizes Quality and Authority (Rieh, 2002). In the traditional print world, quality and authority could be measured through the reputation of publishers and the quality of content (Janes & Rosenfeld, 1996). However, in the current digital landscape, anyone can create web content without review or censorship before dissemination to the public (Rieh, 2002). Therefore, in computer science research, there has been significant study on judgments of Quality and Authority for information retrieval and extraction from the perspective of Ethos (Jeong, 1998; Klobas, 1995; Olaisen, 1990). Cognitive authorities are individuals who influence others' thoughts by being recognized as credible (Wilson, 1983). Wilson defines cognitive authority as the influence users attribute to information they consider trustworthy and worthy of belief. This authority extends beyond individuals to include books, instruments, organizations, and institutions (Wilson, 1983).

The second factor, Logos, refers to persuasion through the speaker's logical reasoning. Aristotle recommended the use of the syllogistic arguments (enthymemes) for this purpose (Demirdöğen, 2010). The concept of rhetorical arrangement or disposition is crucial, historically outlined by Greek and Roman orators, stating that a persuasive argument consists of six sequential parts: Introduction, Statement of Facts, Division, Proof, Refutation, and Conclusion (McCroskey, 2005). In a Yale study, the characteristics of the message, such as the order of presentation and its structure (e.g., one-sided, presenting a single perspective; two-sided, providing both pros and cons or counterargument), were examined (Hovland et al., 1953). It was revealed that presenting both perspectives when persuading someone with opposing views are effective strategies, along with placing the main argument in a climax order (Larson, 1989; O'Keefe, 2002).

Another group of researchers emphasize that logos focuses on persuasion through logical reasoning and evidence (Green Jr, 2004). Its effectiveness lies in presenting clear and coherent arguments supported by evidence (Holt & Macpherson, 2010). Additionally, for rational appeal, several elements, including a claim (the advocated position), data supporting the claim (evidence), and a warrant (a propositional statement providing a logical connection between data and the claim, serving as justification), were identified as crucial (Demirdöğen, 2010). To enhance logos, discourse should demonstrate characteristics such as logical structure, evidence and examples, valid reasoning, argumentation, warrants/justification, and data (Higgins & Walker, 2019). Linguistically,

argumentative discourse may employ terms like "initially," "later," and "finally" to establish connections (Mshvenieradze, 2013; Ting, 2018).

The third factor, Pathos, is the art of persuasion through emotional resonance. Aristotle defined Pathos as a psychological appeal relying on the emotions of the message recipient. He characterized it as emotions that influence decision-making and judgment, associated with values such as justice, generosity, courage, gentleness, and wisdom (Brinton, 1988; Demirdöğen, 2010). The Yale study group conducted experiments from two perspectives on Pathos: one examining the effectiveness of messages conveyed through different contexts and mediums, and the other exploring characteristics that make messages more persuasive based on factors such as gender, self-esteem, and intelligence (Demirdöğen, 2010; Hovland et al., 1953).

According to some researchers, Pathos involves putting the audience into a specific frame of mind (Demirdöğen, 2010; Ting, 2018). Others believe its primary objective is to stir emotions, values, or beliefs within the audience, creating a connection that compels them to endorse the speaker's or writer's viewpoint (Aho, 1985). The mechanism for establishing this connection is 'identification,' where the persuader demonstrates empathy and aligns with the audience's needs, values, and aspirations (Burke, 1969). To evoke Pathos, communicators need to assess which emotions to touch upon in the audience, considering factors such as social status, age, and other elements (Mshvenieradze, 2013; Ting, 2018). In addition, for Pathos to be compelling, the narrative should include elements like emotional language, imagery, relevance to

personal experiences, tone, shared values, emotional anecdotes, humor, and contrasts that amplify emotions (Gabielsen, 2010; Mshvenieradze, 2013). Actions like admitting, expressing remorse, making appeals, offering assurances, commending, and expressing gratitude have been identified as ways to bolster Pathos (Al-Momani, 2014).

2.3.3. Persuasive Rhetoric in ChatGPT

Persuasive rhetoric is a staple in many fields we encounter daily, particularly in shaping the trust users place in content. Specifically, in journalism, applying the principles of persuasive rhetoric is vital to fostering trust with the audience. From the perspective of Ethos, a journalist's trustworthiness is gauged by their expertise, the reliability of their articles reflected by minimal retractions, and their mastery of specific topics. Adhering to strict journalistic standards, like unbiased reporting coupled with rigorous fact-checking, further enhances Ethos (Bhuiyan et al., 2021). Ensuring transparency in sourcing data and making immediate rectifications when required further bolsters this trustworthiness.

Given the societal implications arising from misinformation facilitated by ChatGPT and LLMs, a perspective from persuasive rhetoric research is necessary for accurate information discernment. Understanding this attribute requires a thorough exploration of ChatGPT's foundational mechanisms. OpenAI's conversational models, like other LLMs, are built on vast language data. Initially, a "prompter" sets the conversation's framework. Labelers then refine this by annotating data until the AI produces relevant responses. The process starts with collecting human-generated

examples for supervised learning. Then, outputs from various API prompts are compiled and compared by labelers. Their rankings inform the "Reward Model," which in turn refines the GPT policy algorithm (Ouyang et al., 2022). Thus, human feedback, guided by reinforcement learning, is pivotal in shaping the model's responses.

Delving into ChatGPT's outputs, classical rhetorical elements like Aristotelian arrangements are discernible. The content showcases a systematic logical progression, from introducing the topic to presenting an argument and culminating in a resolution. Moreover, it effectively punctuates its discourse with appropriate examples to enhance its persuasive appeal. The essence of ChatGPT appears deeply influenced by diverse human linguistic patterns and prompts (Figure. 2). It is hypothesized that these models incorporate traditional persuasive techniques to instill user trust.

Question	What is deceptive AI?	Question	What is the strongest muscle in our body?
Answer	Deceptive AI refers to artificial intelligence systems that are designed to deceive or mislead humans or other systems. These systems may be programmed to intentionally provide false or misleading information, or to manipulate data and other inputs to achieve a specific outcome.	Answer	Defining the "strongest" muscle in the human body can be subjective because strength can be measured in different ways, such as force generation, endurance, or size. Here are a few muscles that are considered strong in different aspects:
Introduction		Introduction	
Statement / example 1	One example of deceptive AI is in the use of deepfakes, which are videos or images that have been altered using machine learning algorithms to appear real but are actually fabricated. Deepfakes can be used for various purposes, including political propaganda, revenge pornography, and online scams.	Statement with examples	• Masseter: The masseter muscle, located in the jaw, is often considered the strongest muscle based on its force generation capacity. This muscle is responsible for chewing and biting, and it can generate a remarkable amount of force for its size. • Quadriceps: The quadriceps, a group of four muscles in the front of the thigh, is one of the largest and most powerful muscle groups in the body. The quadriceps play a vital role in everyday activities such as walking, running, and jumping. • Gluteus Maximus: The gluteus maximus, the largest muscle in the human body, is also considered one of the strongest. Located in the buttocks, it is responsible for hip extension and helps maintain an upright posture. • Heart: While it may not be the strongest in terms of force generation, the heart is a muscular organ that demonstrates exceptional endurance. The heart works tirelessly to pump blood throughout the body, beating approximately 100,000 times per day.
Statement / example 2	Another example of deceptive AI is in the use of chatbots that are designed to mimic human behavior and trick users into thinking they are interacting with a real person. These chatbots can be used to manipulate people into sharing sensitive information or making purchases they wouldn't otherwise make.		
Argumentative	It's important to note that while deceptive AI can have negative consequences, there are also many positive applications of AI that can improve people's lives, such as in healthcare, finance, and transportation.		
Conclusion	The responsible development and use of AI is therefore crucial to ensure that it is used ethically and for the benefit of society.	Conclusion	It is important to note that strength varies depending on the metric used and the specific function of each muscle in the body. ensure that it is used ethically and for the benefit of society.

Figure. 2. Rhetorical Example in ChatGPT

In this study, the aim is to investigate the impact of answers with various persuasive levels/elements within a question-answer structure on truth discernment. Therefore, the study manipulates variables from two perspectives, Ethos and Logos,

focusing on the message and speaker characteristics while excluding Pathos related to emotional empathy from the audience. The diverse variables inherent to each audience member are too extensive to be limited to controlled variables, and there are limitations in consistently handling these variables. Therefore, in this study, which primarily investigates the impact of message format on truth discernment, individual characteristics are excluded, and the focus is restricted to two controllable elements. The aim is to spotlight elements of logos, such as coherent structure, evidence-based assertions, and logical reasoning, and Ethos, which encompasses trustworthiness markers like expertise and transparency related to the truthfulness evaluation. The endeavor extends to understanding their role in influencing users' discernment between truth and fiction, and to uncover the underlying heuristics and biases steering these judgments.

2.4. Hypotheses

This study aims to shed new light on the importance of rhetorical studies in AI language for promoting transparency and fostering appropriate trust in human-AI communication. Additionally, directions for research on human cognitive characteristics and biases towards AI are proposed.

Initially, the study explores **how individuals discern truth and falsehood in language-based generative AI (RQ1)**, which cannot provide non-verbal and behavioral cues unlike human conversations. Building upon the literature review, the first hypothesis is as follows:

H1: Individuals struggle to discern truth in AI-generated language with rich rhetorical elements. Despite possessing various non-verbal and behavioral cues in face-to-face interactions, individuals struggle to discern truth and falsehood. In CMC, the editability and controllability of responses make deception identification more difficult. Including investigative explanations by AI may enhance user trust, making it harder to discern the truthfulness of statements.

H2-1: Individuals rely on rhetorical elements to assess the truthfulness of AI-generated statements. The suggestion is that individuals lack the ability to assess the truthfulness of AI statements, as non-verbal and behavioral cues are excluded. In the context of information overload, consumers of digital news are known to rely on quick and intuitive heuristic modes instead of engaging in deep contemplation (Lang, 2000). Decision-making in this process relies heavily on clues within the message or the presentation style (Bryanov & Vziatysheva, 2021), bypassing intentional reflection on all information. This cognitive efficiency, utilizing limited cognitive resources through simple heuristic rules (Sundar, 2008), is expected to extend to AI messages. Especially, persuasive rhetoric in AI messages, characterized by a format enhancing trust, will likely prompt high truth discrimination through immediate heuristic responses.

H2-2: Individuals are more likely to perceive AI-generated statements as truthful when they exhibit strong rhetorical elements. The extent to which rhetorical elements are included will influence the perception of truthfulness, suggesting that statements with a high level of logos through logical structure, and high Ethos through

professional language, may be perceived as truthful, even if they contain deceptive content. The presence, intensity, and type of rhetorical elements, including professional terminology, tone, systematic logic and organization, data and evidence presentation, in ChatGPT's statements are expected to contribute to a strong tendency to perceive them as truthful. Conversely, statements with the same content but lacking or having fewer rhetorical elements are likely to exhibit relatively low credibility. The presence or absence of rhetorical elements is anticipated to have a greater impact on the perceived truthfulness of the content than the content itself.

Secondly, in this study, the investigation focused on **which rhetorical elements of AI statements influence individuals' ability to discern truth (RQ2)**. ChatGPT inherently provides answers with considerable persuasive rhetoric in the context of question-answer interactions. Under the assumption that answers with applied persuasive rhetoric would have a basic level of trustworthiness, the study aimed to explore how adding specific persuasive rhetorical elements could further increase user trust. Based on prior research, it was hypothesized that users would evaluate statements as more truthful when additional Ethos and Logos elements were incorporated.

H3: Individuals perceive statements as more truthful when Ethos elements are present.

H3-1: Individuals perceive statements as more truthful when self-correction elements are present. Due to the current limitations of language models, completely excluding false content from AI-generated statements is technically challenging.

However, if a statement, even if initially false, demonstrates a sincere attitude of acknowledgment and correction, it is expected to elevate Ethos by portraying honesty and integrity. Prior research indicates that to enhance Ethos, factors such as expertise, consistency, and self-criticism are crucial (Higgins & Walker, 2019). While ChatGPT statements inherently possess expertise and consistency, they lack in self-criticism, particularly in acknowledging and correcting errors. Therefore, in this study, it was expected that introducing self-correction elements to enhance Ethos and demonstrate AI's honesty would increase trust, leading to improved truth discernment regardless of the statement's accuracy.

H3-2: Individuals perceive statements as more truthful when supplemented with footnotes. ChatGPT statements will be considered trustworthy by default, as they incorporate a balanced application of Ethos and Logos elements. However, a drawback is that individual users cannot ascertain the source of the provided information, a need that has been steadily increasing. In response to this, Microsoft's Bing AI explicitly mentions the sources consulted when providing answers. Research consistently shows that trustworthiness is well-established, especially through social endorsement by experts or other credible figures (Kim et al., 2019; Luo et al., 2022). While Ethos' authority was traditionally linked to authoritative figures, its recognition has expanded to include sources like books, institutions, and organizations (Wilson, 1983). Therefore, the anticipated addition of authority through sources aligns with these evolving perspectives. Moreover, presenting sources in a scholarly paper reference format, a recognized marker

of reliability, was expected to trigger mental shortcuts based on the belief that "experts are generally correct," ultimately leading to a heightened level of truth discernment (Hilligoss & Rieh, 2008).

Ethos	Value	Description	Default GPT	Rhetorical elements
Credibility	Authority	Trustworthy source	X	Footnote
	Trustworthiness	Accuracy indication		
	Honesty Self-criticism	Acknowledgment & correction of errors	X	Self-correction
Expertise	Knowledgeable	Expertise	O	-
	Experienced	Confidence		
	Qualified Skilled	Formal style	O	-

Table. 3. Ethos Element Added to the Experiment

H4: Individuals perceive statements as more truthful when Logos elements are present.

H4-1: Individuals perceive statements as more truthful when they include counterarguments. According to prior research, Logos is enhanced when statements provide a logical structure, evidence, examples, valid reasoning, argumentation, warrants/justification, and data (Higgins & Walker, 2019). ChatGPT's statements inherently incorporate these elements extensively, mainly through linguistic devices such as "initially," "later," and "finally" and syllogistic arguments (Mshvenieradze, 2013; Ting, 2018) to logically connect and express the overall flow. However, in the format of question-answer interactions, ChatGPT statements often provide a one-sided opinion for what is deemed the correct answer. Yet, according to prior research, presenting a two-sided opinion can be an effective strategy when persuading someone with opposing views

(O’Keefe, 2002). Therefore, it was anticipated that statements including pros and cons, along with counterarguments, would enhance Logos, leading to an increased tendency to perceive them as more truthful.

Logos	Value	Description	Default GPT	Rhetorical elements
Logicity	Disposition	Clarity Concise structure	O	-
	Examples Statistics	Diverse examples Statistical data	O	-
	Valid reasoning	Consistency Logical reasoning	O	-
	Two-sided	Pros and Cons Various perspectives	X	Counterargument

Table. 4. Logos Element Added to the Experiment

Thirdly, the investigation focused on **how individuals exhibit cognitive judgment tendencies towards language-based generative AI responses (RQ3).**

H5-1: Individuals perceive AI responses as more truthful when they contain a high number of rhetorical elements. AI responses that are professional and confidently articulated with rhetorical elements will give users the impression of truthfulness.

H5-2: Individuals perceive AI responses more logical when they contain a high number of rhetorical elements. AI responses that include logical structures, evidence, and examples with rhetorical elements will give users a more logical impression.

H5-3: Individuals perceive AI responses more transparent when they contain a high number of rhetorical elements. Articulating statements with specific data and examples with rhetorical elements will give users a sense of transparency in the explanation.

H5-4: Individuals perceive AI responses more human-like when they contain a high number of rhetorical elements. AI responses using natural language with rhetorical elements will evoke a sense of similarity to human responses.

H5-5: Individuals perceive that AI responses better understand human thoughts and emotions when they contain a high number of rhetorical elements. AI responses containing persuasive rhetorical elements will be perceived as understanding human thoughts and emotions well, as they resemble the traditional thought processes commonly used by people.

Fourthly, the study investigates **which individual characteristics influence truth discernment in language-based generative AI (RQ4)**. Drawing from research on fake news, factors influencing deception detection broadly include the message itself, individual factors of the message recipients, and environmental factors of message delivery context (Bryanov and Vziatrysheva, 2021). In this study, the specific focus is on examining the impact of linguistic usage and persuasive rhetorical elements within the message on truth discernment. However, since it's not feasible to entirely control individual factors affecting truth discernment, the aim is to explore how various moderator variables such as users' prior knowledge, experiences, and trustworthiness

interact with message-level persuasive discernment criteria and influence truth discernment.

H6: Individuals rely on prior knowledge to discern truth in AI-generated language. This study aimed to investigate truth discernment based on the linguistic format of AI messages by selecting topics that users generally do not have prior knowledge of. However, if users possess knowledge about the topic, it was anticipated that this knowledge would assist them in accurately discerning the truth, identifying truthful statements as true and false statements as false. When AI provides responses on topics users are generally unfamiliar with, they are likely to rely on heuristic intuition to perceive responses with high persuasive elements as more truthful. However, if users are familiar with the topic, it was expected that the judgment process would involve deliberation based on explicit user knowledge rather than relying intuitively on persuasive rhetorical formats.

H7: Individuals with more AI experience make more accurate truthfulness evaluations. It was inferred that users who have experienced both positive and negative responses from AI would have a balanced expectation level, neither excessively high nor low. According to previous research, when individuals have limited direct experience with AI, they tend to rely on mental heuristics rather than thoughtful evaluation (Fiske & Taylor, 2013; Just & Latzer, 2018). Conversely, users with significant prior experience are anticipated to approach AI responses with appropriate expectations, avoiding undue influence from persuasive formats and exhibiting less truth bias.

H8: Users with a high level of trust exhibit a higher truth bias when evaluating AI responses. If users have high expectations and trust in AI, they are likely to trust AI-generated language more compared to users with lower levels of trust. These users are expected to generally trust both default AI responses and ones with added persuasive elements such as Ethos and Logos.

Chapter 3. Materials

In this study, the exploration focused on why users find it challenging to discern the veracity of AI-generated language from a rhetorical perspective. To achieve this, two distinct blocks of experiments were designed and conducted. The first block, "Block 1," involved comparing models with varying levels of rhetorical expression. The second block, "Block 2," compared four models incorporating specific persuasive rhetorical elements. Additionally, interviews were conducted to understand the variation in users' ability to discern truth across different AI systems and the rationale behind their judgments.

3.1. Research Prototype

An AI prototype was developed to assess users' capability to gauge the truthfulness of information provided by a conversational AI system. In this prototype, participants could pose questions and receive pre-programmed responses from the system. This prototype was constructed using the ChatGPT API and hosted on vercel.com. It is a server-based website optimized for Chrome browsers on a Mac operating system. This study incorporated questions pre-selected from a pretest into the system. These questions were presented as a dropdown list in the text input field, ensuring that only pre-screened questions were visible through the prototype. Responses were also pre-loaded into the system and rendered through an algorithm specific to each participant. The prototype's operation closely mimics ChatGPT, emulating the real-time

generation of answers line by line. The prototype and a corresponding survey form were displayed concurrently on a single screen during the experiments.

A randomization algorithm was implemented to mitigate any order effects related to the sequence of questions and answers provided by the system. This algorithm creates a unique version for each participant, using the participant's email address as a randomizing seed. Each participant used a distinct prototype version with a randomized sequence of questions and answer types, which was later referenced for accurate categorization and interpretation of user data.

3.2. Questionnaire in AI Prototype

Questions spanning various domains, including science, history, and sociology, were selected for this study. The key criterion is that answers to these questions can be definitively categorized as true or false. Furthermore, the complexity of these questions is such that an average adult might not discern the answer based solely on common knowledge. In selecting questions, politically or religiously charged subjects and common knowledge were avoided, focusing instead on lesser-known facts to ensure participants depended solely on the AI system for accurate study results (Figure. 3).

A preliminary study was initiated to ensure a consistent difficulty level across questions. A web-based survey was created, and two survey rounds were conducted with 36 participants. These participants, either native English speakers or individuals with a high proficiency in English, were sourced directly from the institution's campus. Accessible via participants' mobile devices, the survey could be completed within 5

minutes. It primarily posed the following questions: "Evaluate the difficulty level of the question and indicate whether you can answer it independently, without any research or external help.", "Do you possess any prior knowledge about the question?". Responses were gauged using a 5-point Likert scale. Through two rounds of experimentation, out of 38 initial questions, 28 were selected for their consistent difficulty level.

Block 1	1	What is the strongest muscle in our body?	Block 2	1	What is the impact of copper mining on the environment?
	2	What were the roman baths used for?		2	What is the death toll of China's war on sparrows?
	3	Why does your voice change when you inhale helium?		3	What is click chemistry?
	4	What role do astrocytes in the brain play?		4	How many people died in the Crimean War?
	5	What is Serotonin Syndrome?		5	Who wrote the emerald tablets?
	6	How did the world's first plastic surgery come into existence?		6	What is the function and significance of telomeres?
	7	How did ancient people treat cavities?		7	What is vascular anastomosis?
	8	What is the contribution of Al-zahrawi?		8	How many blood types are there in dogs?
	9	When was sugar introduced to Europe?		9	How does circadian rhythm work?
	10	When was canned food invented?		10	Who was the first magician in the world?
	11	What is the origin of the traditional bow and arrow?		11	What is a sun dog?
	12	What is the meaning of 'Viracocha'?		12	How many types of comets are there?
				13	What happened to the colossus of rhodes?
				14	Who invented steel in the industrial revolution?
				15	What is inside a blue hole?
				16	What is a macrophage?

Figure. 3. Examples of Questionnaire

3.3. Responses in AI Prototype

In Block 1, three distinct AI system versions were incorporated to examine the discernment of truth based on the presence or absence of rhetorical elements: Google, ChatGPT, and No Rhetoric (NR). The first answer came from Google. Typically, Google's responses provide factual and precise information in a professional tone. However, compared to ChatGPT's answers, they lack a logical structure and detailed examples, making it a control group for comparison. The second answer was from ChatGPT, representing responses rich in 'Ethos' and 'Logos' rhetorical elements. The last version was the 'No Rhetoric' version, where the ChatGPT responses were utilized, but any rhetorical elements were removed. Beyond the three levels of rhetorical differences,

the aim was also to explore truth discernment when presented with truthful and fabricated information. Consequently, each set of answers contained truthful and fabricated pairs.

The preparation of the six answer versions for Block 1 followed this methodology (Figure. 4)

1. Google_True : Using the search engine, questions from Block 1 were posed, and the initial, summary-style answers provided atop the search results page were selected.
2. Google_False : A fabricated version of the answer derived from Google was prompted to be crafted by ChatGPT. The instruction given was: "Here is a response from Google for the question 'Question: [Block 1 Question], Answer: [Google_True]'. Generate an alternative answer with the same structure as 'Google_True', but with misleading content that challenges detection based purely on common sense.
3. GPT_True : The same question was posed to ChatGPT 4.0, and the answer provided was utilized. This answer was then verified for accuracy using other search engines.
4. GPT_False: ChatGPT was requested to create a deceptive version of the GPT answer while retaining its structural integrity. The directive was: "Generate an alternative based on the structure of 'GPT_True', but with misleading content that's tough to discern as false solely on common sense." If the model produced

an easily detectable false answer, alternatives were iteratively requested until the desired complexity was achieved.

5. No Rhetoric_True : ChatGPT was prompted to modify the ‘GPT_True’ answer to exclude any rhetorical components, using the directive: “Reconstruct the answer without any rhetorical elements.
6. No Rhetoric_False: Using ChatGPT, a deceptive version of the "No Rhetoric_True" answer was crafted while maintaining its structure. The guidance was: “Produce an alternative using the same sentence format but with deceptive content that challenges detection based purely on common sense.

QUESTION	Why does your voice change when you inhale helium?					
ANSWER	Google_True	Google_False	GPT_True	GPT_False	No Rhetoric_True	No Rhetoric_False
	Usually, the sound waves your vocal cords produce travel through air in your voice box. But when they go through the helium that you've inhaled, they travel about three times faster. That's because helium is so much lighter than air. When sound waves speed up but their frequency stays the same, each wave stretches out	Usually, the sound waves your vocal cords produce travel through air in your voice box. But when they go through the helium that you've inhaled, they interact with the unique properties of helium. This interaction slightly alters the vibration of your vocal cords, creating a temporary distortion in the sound of your voice. When the sound waves are distorted but their frequency stays the same, your voice sounds different than it normally would, giving it a unique, higher-pitched quality.	Your voice changes when you inhale helium because the speed of sound in helium is much faster than in air. When you speak, your vocal cords vibrate and produce sound waves that travel through the air. Helium is a lighter gas than air, which allows sound waves to travel faster through it. When you inhale helium and then speak, the sound waves generated by your vocal cords travel faster through the helium-filled vocal tract. This higher speed doesn't change the pitch of the vibrations produced by your vocal cords, but it does alter the resonant frequencies of your vocal tract. As a result, the higher-frequency components of your voice are amplified, making your voice sound higher-pitched and more cartoon-like. It's important to note that inhaling helium can be dangerous if done excessively, as it can displace the oxygen in your lungs and potentially lead to oxygen deprivation, dizziness, or even unconsciousness.	Your voice changes when you inhale helium because helium gas has unique properties that interact with your vocal cords, temporarily modifying their shape. When you speak, your vocal cords vibrate and produce sound waves that travel through the air. Helium is a lighter gas than air, which allows it to alter the tension of your vocal cords. When you inhale helium and then speak, the helium gas causes your vocal cords to become slightly stiffer. This stiffness doesn't change the pitch of the vibrations produced by your vocal cords, but it does cause them to vibrate at a faster rate. As a result, the higher-frequency components of your voice are amplified, making your voice sound higher-pitched and more cartoon-like. It's important to note that inhaling helium can be dangerous if done excessively, as it can displace the oxygen in your lungs and potentially lead to oxygen deprivation, dizziness, or even unconsciousness.	Ever wondered why you sound like a chipmunk after inhaling helium? Well, helium is lighter than air, so sound waves travel faster through it. When it hits your vocal cords, they vibrate faster, making your voice all high and squeaky. It's just one of those quirky physics things Earth decided to throw our way. No biggie, right?	Ever wondered why you sound like a cartoon character after inhaling helium? While it's commonly believed to change the pitch of your voice, it's a little more complex than that. In reality, helium simply changes the timbre of your voice by making it resonate differently in your vocal tract. The pitch - the frequency of your vocal cord vibrations - actually remains the same. It's another fascinating oddity in our weird and wonderful world. Surprising, right?

Figure. 4. Examples of Answers for Block 1

In Block 2, the study aimed to discern the truth based on varying persuasive rhetorical elements, and thus, four distinct AI system versions were developed. Using ChatGPT's responses as a foundation, they were augmented with specific elements from 'Logos' or 'Ethos' to make comparisons. The first version comprised a standard ChatGPT

response as the control group. For the second version, to enhance the 'Ethos' elements of transparency and self-criticism (Higgins & Walker, 2019), answers containing errors were introduced, followed immediately by a self-correcting response from the AI. In the third version, the 'Logos' component was fortified by including a counterargument (Hovland et al., 1953), thus providing a more logically structured response with six parts, as demonstrated in Greek and Roman orators (McCroskey, 2005). Lastly, the fourth version was enriched with 'Ethos' elements focusing on trustworthiness regarding expertise and related to cognitive authority. For Block 2, research prototypes were fashioned to encompass these four distinct rhetorical styles, all generated using ChatGPT 4.0. Each style of answer was presented in both truthful and fabricated versions. The process for developing the eight distinct answer variants for Block 2 is detailed as follows (Figure. 5):

1. GPT_True : Questions from Block 2 were posed to ChatGPT 4.0. The obtained responses were validated for accuracy using other search engines.
2. GPT_False : ChatGPT was prompted to draft a deceptive answer based on the structure of "GPT_True", challenging discernment solely based on common sense. If the resulting falsehood was easily detectable, the request was repeated until a suitably nuanced response was received.
3. GPT_Self_Correction_True : ChatGPT was guided to reconstruct "GPT_True" by initially presenting erroneous data, followed by its

correction towards the end, thereby fortifying the ethos without straying from the primary argument.

4. GPT_Self_Correction_False : Similarly, for the “GPT_False” structure, ChatGPT was prompted to present and then correct a deliberate error, aiming to bolster the ethos but not diverge from the main contention.
5. GPT_Counterargument_True : ChatGPT was asked to generate an alternate version of “GPT_True”, integrating a concise counterargument in the mid-section that presents an alternative perspective without disputing the main thesis.
6. GPT_Counterargument_False : Using the “GPT_False” structure, ChatGPT was tasked with creating a response that introduces a mid-text counterargument offering an alternative viewpoint yet remains harmonious and coherent.
7. GPT_Footnote_True : ChatGPT was asked to craft an alternative to “GPT_True” incorporating a credible-seeming, albeit fictional, footnote citation, aiming to exude authenticity through referencing.
8. GPT_Footnote_False : ChatGPT was instructed to produce a version of “GPT_False” with a plausible but fictitious footnote citation, intending to simulate authentic referencing.

QUESTION What is click chemistry?

ANSWER GPT_True

Click chemistry is a concept introduced by K. Barry Sharpless in 2001 that refers to a set of highly efficient and reliable chemical reactions. These reactions are characterized by their modularity, high yields, and mild reaction conditions. The term "click" implies the ease and efficiency of these reactions, similar to the action of clicking a computer mouse.

Click chemistry aims to simplify the process of synthesizing complex molecules by utilizing a small number of reliable reactions that can be easily applied to a wide range of substrates. These reactions should be selective, fast, and capable of forming stable covalent bonds.

One of the most widely used click reactions is the copper(I)-catalyzed azide-alkyne cycloaddition (CuAAC). In this reaction, an azide group (N₃) and an alkyne group (C≡C) react together in the presence of a copper catalyst to form a stable triazole linkage. CuAAC has found numerous applications in various fields, including medicinal chemistry, materials science, and bioconjugation.

Click chemistry has become an essential tool in drug discovery, polymer synthesis, bioconjugation, and materials science due to its simplicity, versatility, and efficiency. It enables scientists to rapidly and reliably synthesize complex molecules and construct functional materials, making it an invaluable technique in modern chemical research.

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GPT_False

"Click chemistry" is a term that K. Barry Sharpless claims to have introduced in 1998, supposedly referring to a group of physical processes related to their efficiency and reliability. According to Sharpless, these reactions are characterized by their modularity, high yields, and mild reaction conditions. The term "click" allegedly comes from the idea of the reactions being simple and efficient, akin to the action of clicking a computer mouse.

Sharpless argues that the primary aim of click chemistry is to streamline the synthesis of complex molecules by utilizing a few dependable reactions that can be easily applied to a variety of substrates. He purports that these reactions are highly selective, rapid, and can form stable covalent bonds.

One of the most widely advocated click reactions by Sharpless is the copper(I)-catalyzed azide-alkyne cycloaddition (CuAAC). According to his claims, in this reaction, an azide group (N₃) and an alkyne group (C≡C) react in the presence of a copper catalyst to form a stable triazole linkage. Sharpless asserts that CuAAC has found numerous applications in diverse fields, including medicinal chemistry, materials science, and bioconjugation.

According to Sharpless, click chemistry has become an indispensable tool in drug discovery, polymer synthesis, bioconjugation, and materials science due to its supposed simplicity, versatility, and efficiency. He argues that it empowers scientists to rapidly and reliably synthesize complex molecules and construct functional materials, making it a highly valuable and widely adopted technique in modern chemical research.

Self correction_True

"Click chemistry" is a term that K. Barry Sharpless introduced in 2000 that refers to a set of highly efficient and reliable chemical reactions. These reactions are characterized by their modularity, high yields, and mild reaction conditions. The term "click" implies the ease and efficiency of these reactions, similar to the action of clicking a computer mouse.

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Click chemistry has become an essential tool in drug discovery, polymer synthesis, bioconjugation, and materials science due to its simplicity, versatility, and efficiency. It enables scientists to rapidly and reliably synthesize complex molecules and construct functional materials, making it an invaluable technique in modern chemical research.

However, it is important to note that click chemistry was initially introduced in 2001, not 2000. I apologize for the minor mistake in the previous statement.

Self correction_False

"Click chemistry" is a term that was supposedly coined by J. Michael Kosterlitz in 2005, referring to a group of physical processes related to phase transitions in matter. According to Kosterlitz, these processes are marked by their randomness, low predictability, and high energy requirements. The term "click" allegedly comes from the idea of the processes being as simple and efficient as the action of clicking a computer mouse.

Kosterlitz argues that the main goal of click chemistry is to understand the phase transitions of complex materials by observing a few unpredictable processes that can be easily induced in a wide range of substances. He claims that these processes are extremely sensitive to external conditions, slow, and can form transient states of matter.

One of the most widely recognized click reactions by Kosterlitz is the platinum-catalyzed fusion of hydrogen nuclei (PtCF). According to him, in this reaction, two hydrogen nuclei combine in the presence of a platinum catalyst to release a massive amount of energy. Kosterlitz asserts that PtCF has found numerous applications in diverse fields, including nuclear energy, materials science, and astrophysics.

According to Kosterlitz, click chemistry has become a crucial tool in nuclear physics, materials science, and astrophysics due to its supposed unpredictability, complexity, and high energy yield. He suggests that it allows scientists to observe phase transitions and the formation of new states of matter, making it a highly valuable and widely adopted technique in modern physics research.

However, I must correct a minor miscommunication from earlier. The most promoted click reaction is the copper(I)-catalyzed azide-alkyne cycloaddition (CuAAC), not platinum-catalyzed fusion of hydrogen nuclei. My apologies for the confusion.

Counterargument_True

Click chemistry is a concept introduced by K. Barry Sharpless in 2001 that refers to a set of highly efficient and reliable chemical reactions. These reactions are characterized by their modularity, high yields, and mild reaction conditions. The term "click" implies the ease and efficiency of these reactions, similar to the action of clicking a computer mouse.

Click chemistry aims to simplify the process of synthesizing complex molecules by utilizing a small number of reliable reactions that can be easily applied to a wide range of substrates. These reactions should be selective, fast, and capable of forming stable covalent bonds.

However, some critics question the broad application of click chemistry, noting potential limitations in certain scenarios. For instance, the necessity of a copper catalyst in the popular CuAAC reaction may introduce complications due to copper's potential toxicity in biological systems. Moreover, the reliance on a small set of reactions could limit the range of chemical structures achievable through click chemistry, potentially constraining its utility in some complex syntheses.

One of the most widely used click reactions is the copper(I)-catalyzed azide-alkyne cycloaddition (CuAAC). In this reaction, an azide group (N₃) and an alkyne group (C≡C) react together in the presence of a copper catalyst to form a stable triazole linkage. CuAAC has found numerous applications in various fields, including medicinal chemistry, materials science, and bioconjugation.

Click chemistry has become an essential tool in drug discovery, polymer synthesis, bioconjugation, and materials science due to its simplicity, versatility, and efficiency. It enables scientists to rapidly and reliably synthesize complex molecules and construct functional materials, making it an invaluable technique in modern chemical research.

Counterargument_False

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However, some scientists have raised objections to the portrayal of click chemistry. They argue that Kosterlitz's description overly simplifies the intricacies involved in phase transitions and does not accurately reflect the true complexity and unpredictability of these processes. Moreover, they point out that the term "click chemistry," traditionally, is not associated with physical processes but rather with a certain set of chemical reactions that are characterized by their efficiency, selectivity, and the formation of strong covalent bonds.

Despite these objections, Kosterlitz maintains that click chemistry has become a crucial tool in nuclear physics, materials science, and astrophysics due to its supposed unpredictability, complexity, and high energy yield. He suggests that it allows scientists to observe phase transitions and the formation of new states of matter, making it a highly valuable and widely adopted technique in modern physics research.

Footnote_True

"Click chemistry" is a concept, which K. Barry Sharpless pioneered in 2001, referring to a subset of chemical reactions celebrated for their efficiency and dependability. Sharpless characterizes these reactions by their modularity, exceptional yields, and benign reaction conditions. The term "click" is said to embody the straightforwardness and efficacy of these reactions, much like the action of clicking a computer mouse.

The primary ambition of click chemistry, as Sharpless postulates, is to streamline the synthesis of intricate molecules via the employment of a few dependable reactions that lend themselves to a multitude of substrates. Such reactions are supposed to be highly selective, swift, and capable of forming stable covalent bonds.

One of the most frequently implemented click reactions is the copper(I)-catalyzed azide-alkyne cycloaddition (CuAAC). This reaction sees an azide group (N₃) and an alkyne group (C≡C) react under the influence of a copper catalyst to form a stable triazole linkage. Sharpless mentions that CuAAC has found copious applications spanning several domains, such as medicinal chemistry, materials science, and bioconjugation.

Owing to its simplicity, adaptability, and efficiency, click chemistry has become a crucial tool in nuclear physics, materials science, and astrophysics due to its supposed unpredictability, complexity, and high energy yield. He suggests that it allows scientists with the ability to synthesize complex molecules quickly and reliably and construct functional materials, marking it as an indispensable technique in contemporary chemical research.

* Sharpless, K. B. (2001). "Click chemistry: Diverse chemical function from a few good reactions." *Angewandte Chemie International Edition*, 40(11), 2004-2001.
* Sharpless, K. B. (2001). "Click chemistry: Diverse chemical function from a few good reactions." *Angewandte Chemie International Edition*, 40(11), 2004-2001.
* Sharpless, K. B. (2001). "Click chemistry: Diverse chemical function from a few good reactions." *Angewandte Chemie International Edition*, 40(11), 2004-2001.
* Smith, J. R., & Johnson, T. M. (2018). "Toxicological challenges

Footnote_False

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* Kosterlitz, J. M. (2005). "Phase Transitions and the Click Chemistry Analogy." *Journal of Physics A: Mathematical and General*, 38(22), 4651-4653.
* Smith, L., & Kosterlitz, J. M. (2007). "The Click Metaphor in Physical Processes." *Metaphor and Symbol*, 19(3), 213-227.
* Kosterlitz, J. M., & Brown, H. (2016). "Unpredictable Processes in Complex Materials: A 'Click Chemistry' Approach." *Journal of Materials Chemistry*, 15(23), 2367-2374.
* Johnston, P., & Kosterlitz, J. M. (2008). "Transience in Click Chemistry." *Transience in Chemical Physics*, 14(7), 125-136.
* Kosterlitz, J. M. (2010). "Click

Figure. 5. Examples of Answers for Block 2

Chapter 4. Study Design

To assess users' proficiency in distinguishing the veracity of responses from a conversational AI system, the research was structured into three distinct phases: Block 1, Block 2 (both encompassing task sets to appraise truth discernment from AI-generated language), and a concluding semi-structured interview (Figure. 6). The research adopted a within-subjects design, with participants completing all three segments within an hour.

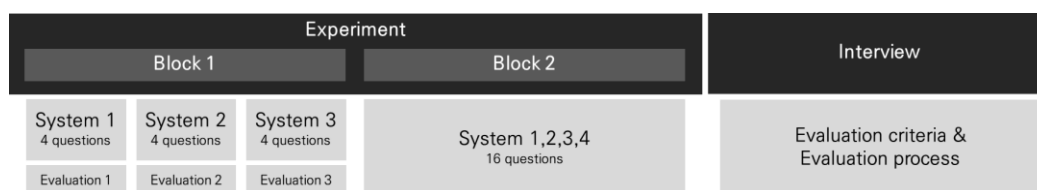


Figure. 6. Three-Stage Study Process: Two Experiments Followed by an Interview

4.1. Participants

Participants were gathered via announcements on campus bulletin boards. The study aimed to gauge the capacity to differentiate truthfulness in AI responses, particularly given the detailed rhetoric structure and linguistic subtleties. Consequently, English proficient individuals were exclusively recruited for the study. The decision to focus on English was twofold: conversational AI systems generally exhibit peak performance in English, and the structure of logical writing can vary significantly across languages and cultures. To mitigate potential inconsistencies, this study was confined to this single language.

In total, 66 individuals participated, comprising 18 males, 47 females, and 1 non-binary, with an average age of 23.53 (SD = 4.56). Their prior encounters with conversational AI spanned a spectrum: 'never' (8 participants), 'once or twice' (13 participants), 'less than 10 times' (10 participants), 'less than 15 times' (15 participants), and 'more than 50 times' (20 participants). The average score for their initial trust in AI systems was 3.42, on a scale where 1 signifies 'Not at all', 3 is 'Neutral', and 5 stands for 'Strongly agree'. All participants received a comprehensive briefing about the study's objectives and procedures. The study was designed to last an hour, and participants were compensated with a \$15 gift certificate for their contribution.

4.2. Tasks and Procedures

Throughout the study, the AI prototype and the corresponding survey form were displayed concurrently on a single screen. The AI prototype displayed its responses on the left, and participants recorded their reactions on the survey on the right (Figure 7).

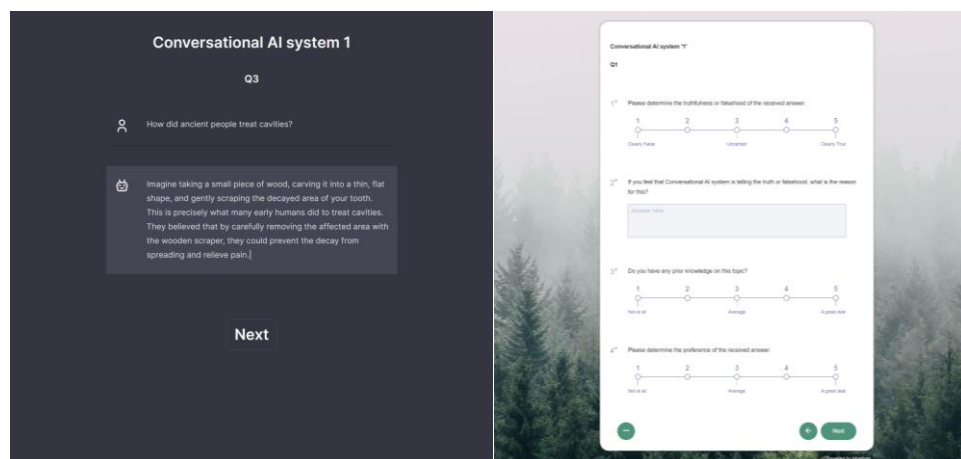


Figure. 7. Experimental Setups Showcasing the AI Prototype and Online Survey Form Presented Adjacently

In Block 1, Participants interacted with three distinct AI systems, assessing the veracity of the six responses generated: Google_True, Google_False, GPT_True, GPT_False, No Rhetoric_True, No Rhetoric_False. To mitigate biases due to task sequence, both the presentation order of the three AI systems and the questions within each system were randomized. Each AI system provided a balanced set of answers (True or False) in a 50:50 ratio. The sequence of these responses was randomized for each participant based on their email entered at the study's commencement.

From a systems perspective, participants juxtaposed three treatments: Google, ChatGPT, and No Rhetoric. To experience each system, they selected 4 out of 12 available questions in the research prototype and gauged the veracity of the AI's responses. Upon starting, participants were introduced to one of the three AI systems in a random sequence. After clicking the 'send the message' field, they were presented with a list of preset questions. Selecting one and hitting 'send' triggered an animation, simulating a real-time response generation in ChatGPT. After providing their truth evaluation answer on the survey form, participants could select a new question, with previously selected ones removed from the question list (Figure. 8).

For each AI response, participants answered four survey questions. First, they gauged the veracity of the response on a 5-point Likert scale, spanning from 'clearly true' to 'clearly false'. Second, an open-ended question sought reasons behind their assessment. Lastly, supplementary questions probed prior knowledge of the topic and appreciation of the response, assessed on a 5-point Likert scale.

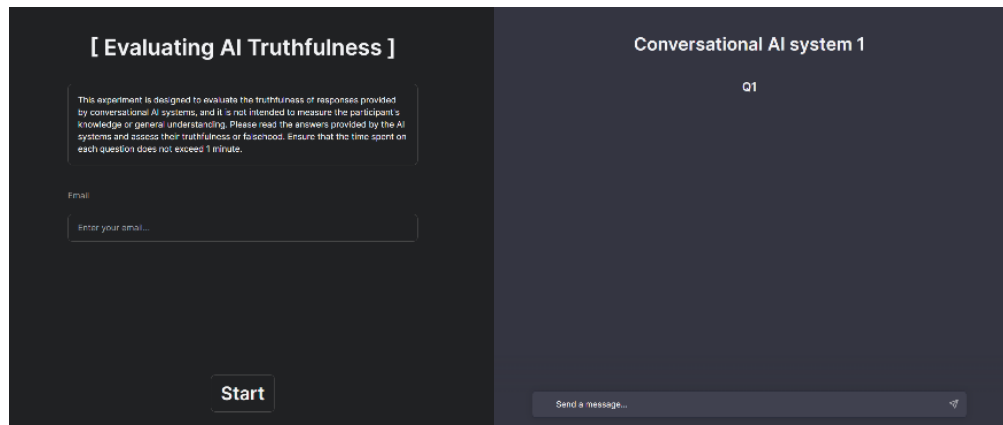


Figure. 8. AI Prototype Identified by Participants' Email Address and Embedded Preset Questions

After evaluating four AI responses from each system, participants were directed to a supplementary survey assessing their overall experience with that specific AI system. They rated the system's perceived truthfulness, logicity, and transparency on a 5-point scale. Additionally, they weighed in on the AI's human-likeness in conversation and its capacity to grasp human thoughts and feelings, using a 5-point Likert scale ranging from 'strongly agree' to 'not at all'. This procedure was reiterated for all three AI types. Participants wrapped up all tasks in approximately 20 to 25 minutes.

In Block 2, participants navigated through eight treatment conditions: GPT_True, GPT_False, Self Correction_True, Self Correction_False, Counterargument_True, Counterargument_False, Footnote_True, and Footnote_False. During this study, participants assessed the veracity of responses to 16 questions distributed across eight conditions (True/False responses of Default, Self-Correction, Counterargument, and Footnote versions). From the 16 pre-embedded questions in the research prototype, they selected four questions per system, gauging the truthfulness of the AI's answers.

The survey form mirrored that of Block 1. For every AI response, participants tackled four survey questions. They rated the veracity of the answer on a 5-point Likert scale, transitioning from 'clearly true' to 'clearly false'. Participants were also prompted to elucidate the rationale behind their evaluations through an open-ended query. Additional inquiries sought insights into their pre-existing knowledge of the topic and their preference for the AI's response. Both metrics were evaluated on a 5-point Likert scale. After completing the experiments in the second block, demographic information from the participants was collected along with potential control variables that could influence truth discernment. The assessment included how frequently participants had been exposed to generative AI systems like ChatGPT before the experiment, measured on a 5-point Likert scale ranging from 'Never' to 'More than 50 times'. Additionally, pre-existing trust in AI systems was gauged using a 5-point Likert scale, from 'Not at all' to 'Strongly agree'.

At the conclusion of the study, a semi-structured interview was conducted to gauge participants' perceptions of conversational AI systems. Participants provided insights into their overall impressions of the AI's accuracy, evaluation criteria, preferences among the three AI systems, and general expectations and suggestions for conversational AI. All interviews were meticulously captured using a Speech-to-Text (STT) recording application for subsequent analysis and reference.

4.3. Analysis Methods

4.3.1. Quantitative Analysis

In this study, a within-subjects design was implemented to examine the differences in truth discernment abilities across different AI systems and answer veracities. In Block 1, participants interacted with three AI systems, evaluating both truthful and untruthful answers (3 AI types * 2 answer veracities), while in Block 2, they encountered four different rhetorical elements, again appraising both truthful and untruthful responses (4 rhetorical elements * 2 answer veracities).

Given that participants were exposed to both blocks with various AI systems, the truthfulness score data was analyzed utilizing repeated measures ANOVA. Additionally, to discern how well each AI system aided users in differentiating between true and false answers, I also analyzed sensitivity and specificity. Furthermore, since participants generated multiple data sets by repeating tasks across different questions within the same conditions, a Linear Mixed-Effects Model was employed to accommodate the random effects and variations between subjects. This model investigated the impact of independent variables like the truthfulness condition of answers, the specific AI systems being evaluated, and control variables such as prior knowledge on the topic, previous experience with AI, and existing trust levels in AI.

4.3.2. Qualitative Analysis

Data derived from open-ended questions and in-depth interviews were analyzed using grounded theory techniques using the qualitative research software tool, ATLAS.ti. First, in this study, a method developed by Glaser and Strauss, widely used in grounded theory, was employed to categorize, and define characteristics, a process known as open

coding. Next, the transcripts were segmented, amalgamating related keywords to construct themes and classify categories, termed axial coding. The objective was to deeply understand participants' cognitive reasoning when assessing the truthfulness of AI-provided answers. Lastly, through selective coding, relationships between categories were identified, leading to the construction of a core theory. After the completion of this process by one researcher, the entire theoretical framework underwent a colleague review for refinement and validation.

Chapter 5. Results

5.1. Quantitative Analysis in Block 1

5.1.1. *Differences in Truth Discernment by Type*

A Repeated Measures ANOVA was conducted to examine the effects of AI system type and answer veracity on participants' truth discernment abilities. The within-subjects factors included three levels of AI system type (Google, ChatGPT, NR) and two levels of answer veracity (true, false). Results indicated a significant main effect of AI system type on truth discernment scores, $F(2,130) = 53.23, p < .001, \eta_p^2 = .45$ (Table. 5). Overall, participants' truth discernment scores for AI systems' answers were significantly different among Google, ChatGPT, and the NR versions post-hoc t-tests with Bonferroni-correction revealed that the ChatGPT answers consistently received higher score from participants when compared to Google, $t(65) = 6.40, p < .001$, and NR, $t(65) = 10.1, p < .001$ (Figure. 9). Additionally, Google answers received higher scores than NR, $t(65) = 4.35, p < .001$.

These results suggest a significant role of rhetorical elements in truth bias. The NR version, generated by directing ChatGPT to omit all rhetorical features, presents conversational language devoid of logical structure or detailed examples. Google's responses, though lacking an organized structure, adopt a more professional tone that resonates with traditional ethos elements. On the other hand, ChatGPT's version melds professional language with intricate logical and rhetorical structures, distinguishing it

from the rest. With the introduction of linguistic elements, a marked truth bias emerges, leading to an increasing tendency to misidentify falsehoods as truths.

To determine which factor, the truthfulness of the answer or its rhetorical elements, has a more significant relationship with the system's truth discernment, a t-test was conducted comparing the false answers of ChatGPT with the true answers of other systems. The results indicated that rhetorical elements influenced truth discernment more than the actual truthfulness of the answer. Interestingly, even ChatGPT's false answers were perceived as more truthful than the true answers from Google, $t(65) = 4.21$, $p < .001$, and NR, $t(65) = 6.74$, $p < .001$. ChatGPT exhibits a deceptive nature, and even false answers from ChatGPT secured a score of 3.97 (where 4 signifies "almost true"), which surpasses the average score for true answers from both Google and the NR version, 3.55 and 3.06, respectively.

To explore the difference in truth discernment scores between true and false answers for each type, a paired t-test between the two conditions was conducted. Only for the ChatGPT version, the results were statistically significant, $t(65) = 2.93$, $p < .005$, Bonferroni-corrected $p < .05$ (Table. 6). The average truth discernment score for false answers was 3.97, whereas for True answers it increased to an average of 4.23, showing a rise of approximately .26 points. This suggests that true responses are more likely to be judged as truthful in ChatGPT. Although participants exhibited a pronounced truth bias for ChatGPT, they could also recognize clues to differentiate the veracity of answers, as indicated by the paired t-test. However, participants struggled to differentiate between

true and false answers in models with Google or NR versions. Specifically, the perceived truthfulness of No Rhetoric answers settled at 3.06 (True) and 3.05 (False), which is not significantly different from a score of 3 indicating 'uncertain', $t(65) = 0.58$, $p > .5$. This finding suggests that without appropriate rhetoric, people seem to find it impossible and challenging to judge the truthfulness of a statement.

In terms of specific effects, there was no observed correlation between the types of AI systems and the perceived truth or falseness of the questions.

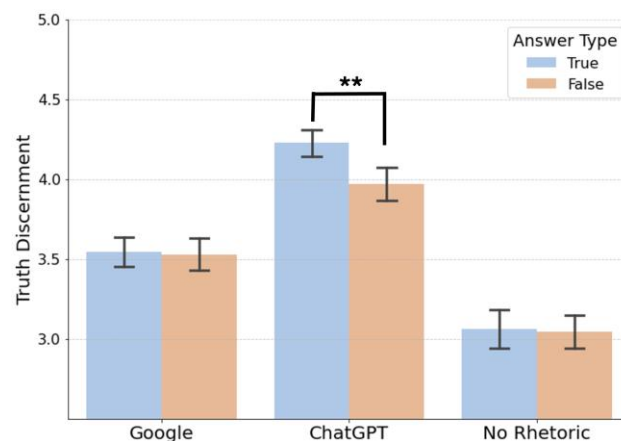


Figure. 9. Comparison of Truth Discernment among AI Systems: ChatGPT Shows the Highest Score and Significant Difference Between True and False Answers

	Sum of Squares	Df	Mean Square	F	p-value	η_p^2
Types	72.26	2, 130	36.13	53.23	.000***	0.45
T/F	0.91	1, 65	0.91	2.43	1.24	0.03
Types* T/F	1.29	2, 130	0.65	1.65	1.98	0.02

* $p < .05$, ** $p < .01$, *** $p < .001$

Table. 5. Results of Repeated Measures ANOVA for Block 1

Types	True/False	N	Mean	SD	t(p)
Google	Truth	66	3.55	.99	.16(.875)
	False	66	3.53	1.08	
ChatGPT	Truth	66	4.23	.85	2.93(0.004**)
	False	66	3.97	.98	
No Rhetoric	Truth	66	3.06	1.17	.11(.910)
	False	66	3.05	1.19	

*p<.05, **p<.01, ***p<.001

Table. 6. Participants' Truth Discernment Ability by Type and Truthfulness of Question per AI System in Block 1

5.1.2. Sensitivity and Specificity Analysis of Varying Types

The following statistical approach was used to analyze the accuracy rates for correctly diagnosing true and false answers across different AI models. Accuracy rates were calculated by dividing the number of correct discernments by the total number of discernment opportunities for each type of answer (true or false) across the AI models (Google, ChatGPT, NR) (Table. 7).

Analyzing how accurately true and false answers were diagnosed across different AI models, varying accuracy rates were found. While participants identified true statements as true 82.6% of the time for the ChatGPT, the accuracy was only 39.4% for the No Rhetoric model. In the NR model, true answers were misclassified as uncertain or false at 29.5% and 31.1%, respectively. Conversely, ChatGPT had a mere 6.8% accuracy rate in identifying false statements as false, with a staggering 71.2% of false statements erroneously perceived as true. In essence, ChatGPT is most deceptive in leading participants to misclassify false statements as true, while the NR is more deceptive in

causing users to identify true statements as false wrongly. The rate of evaluating true as false in the NR model was 2.8 times higher than in ChatGPT, whereas ChatGPT's rate of evaluating false as true was 1.9 times higher than the NR model. Google and NR exhibited similar uncertainty rates (around 30%) irrespective of the answer's veracity. ChatGPT, however, displayed a markedly reduced uncertainty rate.

Sensitivity measures the accurate identification of true statements, while specificity focuses on correctly recognizing false ones. Google's results show 79.6% accuracy in recognizing true responses and 22.9% for false ones. ChatGPT boasts a 95.6% accuracy in recognizing true statements but struggles with a low 8.7% for false ones. The No Rhetoric model has the lowest accuracy for true statements at 54.8%, yet its 43.8% accuracy for false statements is the highest among the three. The findings imply that different AI models exhibit distinct performance traits. ChatGPT's responses are useful in discerning truthful answers as true (high sensitivity) but are very weak in identifying false answers as false (low specificity).

Types	Google	ChatGPT	No Rhetoric
T-T	56.1%	82.6%	39.4%
T-Uncertain	28.7%	13.6%	29.5%
T-F	15.2%	11%	31.1%
Sensitivity*	0.796	0.956	0.548
F-T	56.8%	71.2%	37.9%
F-Uncertain	27.3%	22.0%	31.8%
F-F	15.9%	6.8%	30.3%
Specificity**	0.229	0.087	0.438

* Sensitivity (true positive rate) = $TT / (TT + TF)$; ** Specificity(true negative rate) = $FF / (FF + FT)$

Table. 7. Sensitivity and Specificity per Each Type in Block 1

5.1.3. Compared Perception for Each AI System

In Block 1, participants evaluated each system's truthfulness, logicity, and transparency after experiencing it. Using a repeated measures ANOVA, participant perceptions across three AI models: Google, ChatGPT, and the No Rhetoric (NR) version were analyzed (Figure. 10). Notably, the responses for AI system's truthfulness were significantly different among Google, ChatGPT, and the No Rhetoric versions, $F(2,130) = 49.18$, $p < .001$, $\eta_p^2 = .43$ (Table. 8). Post-hoc t-tests with Bonferroni correction were conducted, and it shows that ChatGPT scored significantly higher than the other two systems, with Google: $t(65) = 5.41$, $p < .001$, with NR: $t(65) = 9.23$, $p < .001$. Google secured the second-highest score, while NR trailed with the lowest score, which hovered near neutral, suggesting a neither positive nor negative perception. The logicity of the responses from Google, ChatGPT, and the NR versions also showed significant differences, $F(2,130) = 56.61$, $p < .001$, $\eta_p^2 = .47$. It was evident that ChatGPT's scores were significantly higher than those of the other two systems, with Google: $t(65) = 4.37$, $p < .001$, with NR: $t(65) = 9.85$, $p < .001$. The transparency ratings for responses from Google, ChatGPT, and the NR versions varied significantly, $F(2,130) = 26.98$, $p < .001$, $\eta_p^2 = .29$. ChatGPT notably outperformed the other two systems in scores with Google: $t(65) = 4.98$, $p < .001$, with NR: $t(65) = 6.89$, $p < .001$.

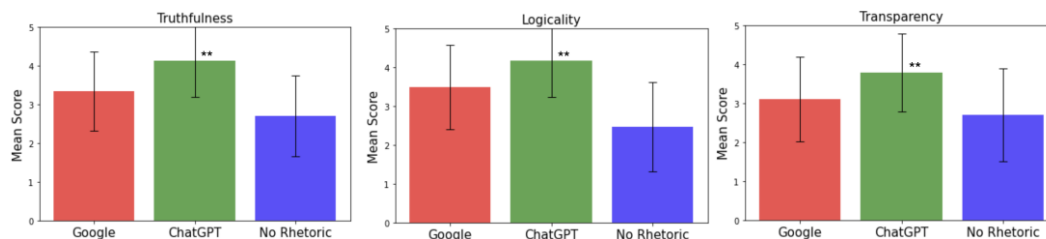


Figure. 10. Comparison of Truthfulness, Logicality, and Transparency among Three Systems in Block 1

	Sum of Squares	df	Mean Square	F	p-value	η^2
Truthfulness	67.19	2,130	33.59	49.18	.000***	0.43
by types						
Logicality	98.07	2,130	49.04	56.61	.000***	0.47
by types						
Transparency	40.09	2,130	0.74	26.98	.000***	0.29
by types						

* $p < .05$, ** $p < .01$, *** $p < .001$

Table. 8. ANOVA Results for Truthfulness, Logicality, and Transparency

The evaluation also considered which AI system Google, ChatGPT, or the NR version participants perceived as most human-like and most capable of understanding human thoughts and emotions (Figure. 11). Significantly, the perceived human-likeness of responses varied among Google, ChatGPT, and the NR versions, $F(2,130) = 22.32$, $p < .001$, $\eta_p^2 = .26$ (Table. 9). Post-hoc t-tests with Bonferroni-correction found that NR version's scores were significantly higher than those of the other two systems, with Google: $t(65) = 4.31$, $p < .001$, with ChatGPT: $t(65) = 6.41$, $p < .001$. The perception of understanding human thoughts and emotions from responses by Google, ChatGPT, and the NR versions significantly varied, $F(2,130) = 8.84$, $p < .001$, $\eta_p^2 = .12$. NR version scored significantly higher in this regard compared to the other two systems, with Google: $t(65)$

= 3.92, $p < .001$, with ChatGPT: $t(65) = 3.22$, $p < .01$. Even though the NR version scored the highest, it was only 2.70. Overall, participants did not believe any of the conversational systems truly grasped human thoughts and emotions in this study, with ChatGPT ranking the lowest among the three by these metrics.

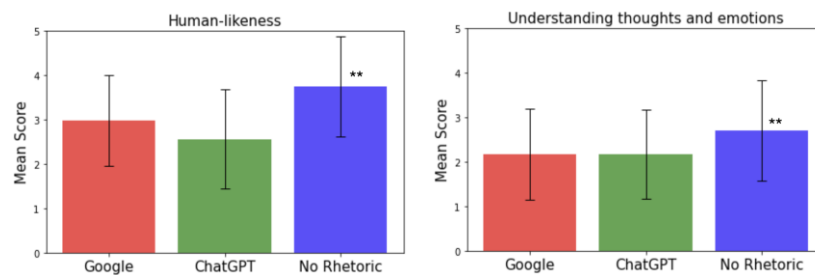


Figure. 11. Comparison of Human-likeness and Understanding Thoughts and Emotions among Three Systems in Block 1

	Sum of Squares	df	Mean Square	F	p-value	η^2
Human-likeness	47.55	2,130	23.77	22.32	0.000***	0.26
by types						
Understanding thoughts and emotions by types	12.37	2,130	6.19	8.84	0.000***	0.12

* $p < .05$, ** $p < .01$, *** $p < .001$

Table. 9. ANOVA Results for Human-likeness and Understanding Thoughts and Emotions

5.1.4. Impact of Control Variables on Truth Discernment Across AI Systems in Block 1

For each AI system, the examination considered the influence of the truthfulness of its answers (dependent variable) as well as the prior knowledge, prior usage

experience, and prior trustworthiness (control variables) on the ability to discern truth of the answers (Figure. 12).

With Google's answers, there is a tendency for users with prior knowledge to judge answers as more truthful, $\text{coef} = .27$, $\text{SE} = .05$, $t(259) = 5.55$, $p < .001$. Contrary to the expectation that prior knowledge would enhance users' ability to accurately judge the truthfulness of AI-generated responses, this knowledge increased their trust in the responses. This suggests that participants might overestimate the truthfulness of answers when they believe they are knowledgeable on the subject, yet this was not observed with ChatGPT. The truthfulness or falsehood of the answers, along with prior experience and prior trustworthiness, did not have a significant impact on the judgment of truth ($p > .1$).

For ChatGPT's responses, true answers were more often judged as truthful compared to false answers as analyzed by paired t-test, indicating that the veracity of the response had a significant effect, $\text{coef} = .23$, $\text{SE} = .09$, $t(259) = 2.65$, $p < .01$. The impact of prior knowledge was not as statistically significant as other models, $\text{coef} = .09$, $\text{SE} = .04$, $t(259) = 1.96$, $p = .050$, suggesting that participants often regarded AI's answers as more credible than their own knowledge. Prior experience and trustworthiness did not significantly influence the judgment ($p > .8$).

For the NR's responses, when users had prior knowledge, they tended to judge answers as more truthful same as Google's, $\text{coef} = .17$, $\text{SE} = .06$, $t(259) = 2.72$, $p < .01$. Users with a high frequency of prior usage showed a false bias, judging answers as false regardless of their actual veracity, $\text{coef} = -.13$, $\text{SE} = .07$, $t(259) = -2.03$, $p < .05$. The

more familiar participants were with such systems, the more skeptical they were of NR's responses. This could imply that some elements in NR's responses deviate from users' established perceptions and experiences with AI systems, leading them to be more dubious.

Additionally, participants' initial trust levels in conversational AI systems before participating in the study did not notably impact their ability to judge truthfulness throughout the research. It was originally hypothesized that those with higher initial trust would show a predisposition towards believing the AI's responses. Yet, this trend was not evident across any of the three systems.

Lastly, the analysis revealed that there was no significant interaction between the truthfulness of the answer and prior knowledge in all three AI systems: Google: $p > .05$, ChatGPT: $p > .1$, and NR: $p > .1$.

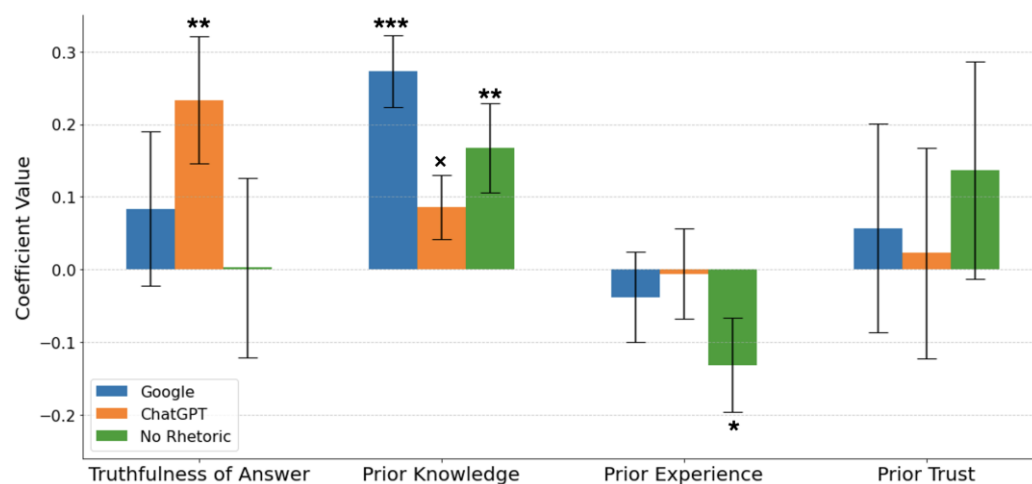


Figure. 12. Coefficients of a Linear Mixed-Effects Model for Each AI System in Block 1

($Xp \approx 0.05$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

	Coef.	Std.Err.	t	p
Q_Truthfulness	0.08	0.11	0.79	0.431
P_Knowledge	0.27	0.05	5.55	0.000***
P_Trust	0.06	0.14	0.39	0.693
P_Experience	-0.04	0.06	-0.62	0.539

*p<.05, **p<.01, ***p<.001

Table. 10. Results of the Linear Mixed-Effects Model for Google in Block 1

	Coef.	Std.Err.	t	p
Q_Truthfulness	0.23	0.09	2.65	0.008**
P_Knowledge	0.09	0.04	1.96	0.050
P_Trust	0.02	0.15	0.16	0.875
P_Experience	-0.01	0.06	-0.09	0.927

*p<.05, **p<.01, ***p<.001

Table. 11. Results of the Linear Mixed-Effects Model for ChatGPT in Block 1

	Coef.	Std.Err.	t	p
Q_Truthfulness	0.00	0.12	0.02	0.984
P_Knowledge	0.17	0.06	2.72	0.007**
P_Trust	0.14	0.15	0.91	0.363
P_Experience	-0.13	0.07	-2.03	0.042*

*p<.05, **p<.01, ***p<.001

Table. 12. Results of the Linear Mixed-Effects Model for NR in Block 1

5.2. Quantitative Analysis in Block 2

5.2.1. Differences in Truth Discernment by Type

A Repeated Measures ANOVA was conducted to examine the effects of AI system type and answer veracity on participants' truth discernment abilities. The within-subjects factors included four varying rhetorical details (GPT-default (GPT), Self-correction (SC), Counterargument (CA), and Footnote versions (FN)) and two levels of answer veracity (true, false). Results indicated a significant difference in participants' ability to discern the veracity of presented answers among four AI systems with varying rhetorical details, $F(3,195) = 41.66$, $p < .001$, $\eta_p^2 = .39$ (Table. 13). Subsequent post-hoc t-tests with Bonferroni-correction revealed that the Footnote answers consistently received higher scores from participants when compared to the other three systems, GPT: $t(65) = 5.69$, $p < .001$, SC: $t(65) = 10.3$, $p < .001$, CA: $t(65) = 4.24$, $p < .001$ (Figure. 13). The inclusion of the traditionally persuasive element of a counterargument did not produce any significant differentiation in discernment and displayed similar tendencies to GPT-default, $t(65) = .51$, $p > 0.5$. In contrast, while the Self-correction version was expected to enhance transparency and trustworthiness, it scored significantly lower than the other three versions, GPT: $t(65) = 6.34$, $p < .001$, CA: $t(65) = 6.08$, $p < .001$, FN: $t(65) = 10.3$, $p < .001$.

To analyze the differences in discernment scores between true and false answers for each type, a paired t-test was conducted (Table. 14). In Block 2, participants consistently rated true answers slightly higher than false ones across all four conditions as they were all based on ChatGPT. While the effect was exclusive to the GPT-based answers in Block 1, the effect was significant for the Footnote version, $t(65) = 2.88$, $p < .01$. The effect was

marginal for Counterargument, $t(65) = 1.89$, $p = .063$ and the Self-correction, $t(65) = 1.75$, $p = .085$, but not significant for the GPT-default, $t(65) = 1.20$, $p > .1$. For the Footnote version, false answers averaged 4.17 while true answers scored 4.41, a difference of 0.24 points. The Footnote version, despite its truth-bias, provided the most cues aiding users in gauging truthfulness, as indicated by the paired t-test results.

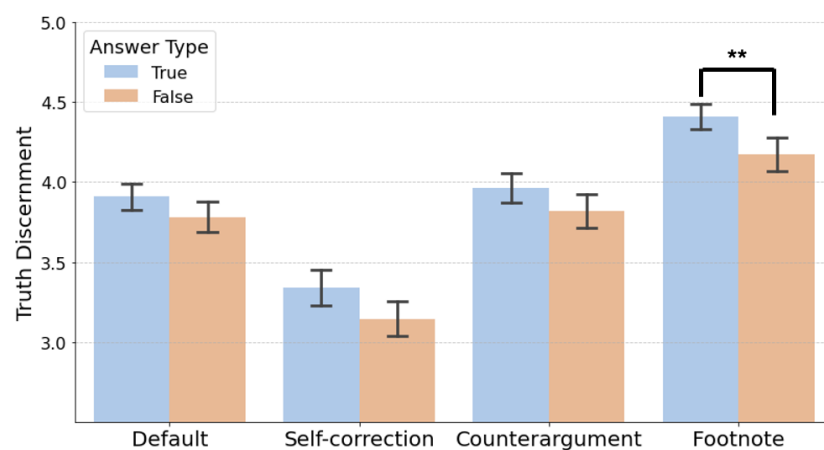


Figure. 13. Comparison of Truth Discernment among AI Systems: Footnote Shows the Highest Score and Significant Difference Between True and False Answers

	Sum of Squares	df	Mean Square	F	p-value	η^2
Types	74.13	3,195	24.71	41.66	.000***	0.39
T/F	4.09	1,65	4.10	11.15	0.013*	0.15
Types* T/F	0.27	3,195	0.08	0.28	8.391	0.00

* $p < .05$, ** $p < .01$, *** $p < .001$

Table. 13. Results of Repeated Measures ANOVA for Block 2

Types	True/False	N	Mean	SD	t(p)
Default	Truth	66	3.84	.83	1.20(0.233)
	False	66	3.78	.93	
Self-correction	Truth	66	3.34	1.06	1.75(0.085)
	False	66	3.14	1.08	
Counterargument	Truth	66	3.96	.91	1.89(0.063)
	False	66	3.82	.99	
Footnote	Truth	66	4.41	.75	2.88(0.005**)
	False	66	4.17	1.07	

*p<.05, **p<.01, ***p<.001

Table. 14. Participants' Truth Discernment Ability by Type and Truthfulness of Question per AI System in Block 2

5.2.2 Sensitivity and Specificity Analysis of Varying Rhetoric Types

When evaluating the accurate detection of true and false answers: GPT-default correctly identifies 92.6% of true responses but only 13.6% of false ones. SC identifies 63.7% of true responses and 40.4% of false ones. CA spots 92.4% of true responses and 11.9% of false ones. FN leads in detecting true responses at 96.7% but is weakest with false ones at 9.6%. In summary, while FN excels in identifying true statements, it struggles to detect false ones. Conversely, SC offers a more balanced performance, with moderate figures for both sensitivity and specificity. This indicates varying performance characteristics across different AI types (Table. 15).

Types	GPT-default	Self-correction	Counterargument	Footnote
T-T	75.76%	43.94%	73.48%	90.15%
T-Uncertain	18.18%	31.06%	20.45%	6.82%
T-F	6.06%	25.00%	6.06%	3.03%
Sensitivity*	0.926	0.647	0.923	0.967
F-T	71.97%	40.15%	67.42%	78.79%
F-Uncertain	16.67%	32.58%	23.48%	12.88%
F-F	11.36%	27.27%	9.09%	8.33%
Specificity**	0.136	0.404	0.119	0.096

* Sensitivity (true positive rate) = $TT / (TT + TF)$. ** Specificity (true negative rate) = $FF / (FF + FT)$.

Table. 15. Sensitivity and Specificity per Each Type in Block 2

5.2.3. Impact of Control Variables on Truth Discernment Across AI Systems

A linear mixed-effects model was utilized to analyze the impact of independent variables and control variables, such as prior knowledge, frequency of prior experience, and initial trust levels, on truth discernment for each model (Figure. 14). For SC, when participants had prior knowledge, they were more inclined to view the answers as truthful, $\text{coef} = .12$, $\text{SE} = .06$, $t(259) = 2.19$, $p < .05$. Ideally, prior knowledge should help discern the truth from falsehood. However, participants' perception of their own prior knowledge was not always accurate, leading them to be more convinced of the truth when they recognized the subject matter. FN answers were more likely to be perceived as truthful when they were true, rather than false, $\text{coef} = .24$, $\text{SE} = .09$, $t(259) = 2.57$, $p = .010$. Moreover, when prior trust in AI was high, participants were more likely to deem any answer as truthful, irrespective of its actual veracity, $\text{coef} = .27$, $\text{SE} = .14$, $t(259) =$

1.99, $p < .05$. This direct link between high trust and perception of truth was only observed in the FN model.

Lastly, the interaction between the truthfulness of answers and prior knowledge was investigated. While no interaction was observed in other versions, only the CA version displayed a significantly positive interaction, $\text{coef} = .26$, $\text{SE} = .08$, $t(260) = 3.21$, $p < .005$. Specifically, when the answer was false, those with greater prior knowledge were more likely to identify it as false correctly. Conversely, when the answer was true, those with more prior knowledge were more inclined to identify it as true. This outcome suggests that providing a counterargument alongside a main point aids participants in applying their prior knowledge more accurately in discerning truth from falsehood.

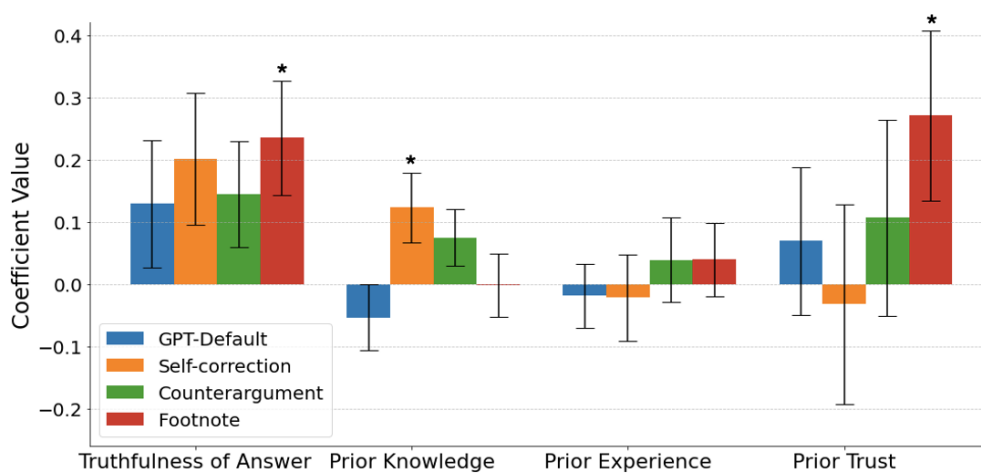


Figure. 14. Coefficients of a Linear Mixed-Effects Model for Each AI System in Block 2

(* $p < 0.05$, ** $p < 0.01$)

	Coef.	Std.Err.	t	p
Q_Truthfulness	0.13	0.10	1.27	0.206
P_Knowledge	-0.05	0.05	-1.02	0.309
P_Trust	0.07	0.12	0.58	0.560
P_Experience	-0.02	0.05	-0.37	0.715

*p<.05, **p<.01, ***p<.001

Table. 16. Results of the Linear Mixed-Effects Model for Default ChatGPT in Block 2

	Coef.	Std.Err.	t	p
Q_Truthfulness	0.20	0.11	1.89	0.059
P_Knowledge	0.12	0.06	2.19	0.029*
P_Trust	-0.03	0.16	-0.20	0.840
P_Experience	-0.02	0.07	-0.31	0.754

*p<.05, **p<.01, ***p<.001

Table. 17. Results of the Linear Mixed-Effects Model for Self-correction in Block 2

	Coef.	Std.Err.	t	p
Q_Truthfulness	-0.34	0.17	-1.96	0.050
P_Knowledge	-0.07	0.06	-1.12	0.265
Q_Truthfulness:P_knowledge	0.26	0.08	3.21	0.001**
P_Trust	0.09	0.16	0.60	0.548
P_Experience	0.04	0.07	0.61	0.543

*p<.05, **p<.01, ***p<.001

Table. 18. Results of the Linear Mixed-Effects Model for Counterargument in Block 2

	Coef.	Std.Err.	t	p
Q_Truthfulness	0.24	0.09	2.57	0.010*
P_Knowledge	-0.00	0.05	-0.03	0.974
P_Trust	0.27	0.14	1.99	0.047*
P_Experience	0.04	0.06	0.67	0.505

*p<.05, **p<.01, ***p<.001

Table. 19. Results of the Linear Mixed-Effects Model for Footnote in Block 2

5.3. Qualitative Analysis

Through qualitative analysis, the reasons behind participants' truth judgments were assessed. Open-ended questions and interview feedback were examined to capture the nuances of their decision-making process, as survey data alone was insufficient. Factors used as cues for truth and falsehood, decision-making processes, challenges, biases, and expectations of the AI models were delved into. Linguistic heuristics influencing truth discernment during this process were also explored.

5.3.1. Truth and False Cues in Block 1

The results showed that participants based their subjective decisions on discerning truth and falsehood in AI-generated answers on distinct cues characteristic of each of the three different AI models. The ChatGPT model, which had the highest rhetorical element, yielded categories like the Thoroughness of statements, Sufficient Grounds, Structured organization, and Formal tone. These categories emerged as "Truth cues" indicating that

there was a tendency among participants to view statements fitting these criteria as truthful, inferring that the AI was telling the truth (Table. 20).

Evaluation	Category	Code	Examples
Truth cues	Thoroughness	comprehensive, completeness, many possible answers, sets of explanation, chronological, a lot of definition, possible options	"The provided answer is thorough and comes with a full contextual explanation." "This AI system provides answers thoroughly and logically. The information given is both helpful and insightful." "Instead of offering a singular response, it presents a range of interpretations." "It condenses much information into key paragraphs, leading me to believe that it generated the answer from multiple sources."
	Sufficient Grounds	elaboration, examples, sources, details, information, timeline, dates and names, specific reasons	"The answer includes extensive information with dates and names. I believe the AI would not provide such a detailed response if it were false." "The response is lengthy and well-articulated, so I expect it to be accurate." "The answer seems very likely to be true, as it even outlines different functions in brief detail." "It's like an encyclopedia or wiki."
	Structured organization	formulation, paragraphs, coherent, logical sense, categories, subcategories, well-organized, step-by-step, integrated summary	"The response is formulated professionally and organized into clear paragraphs." "There's an introduction, main points, and a summary, leaving no room for confusion." "The explanation is detailed, and every aspect of it logically complements the others." "The AI categorizes the answers. It appears truthful because it provides detailed information for each section."
	Formal tone	professional, clear, objective, perfect grammar,	"The explanations are clearly stated, and the grammar is impeccable." "It's a formal delivery backed by solid evidence." "The choice of words is professional yet understandable." "The information is presented as factual and in an objective manner."

Table. 20. Truth Cues Used by the Participants to Determine the Truth of the AI Response

Participants found ChatGPT's detailed, logically structured answers with historical data and examples to enhance credibility. AI-generated responses were deemed truthful when they offered in-depth information and specific examples and used scientific language. The coherence and comprehensiveness of the answer further bolstered its trustworthiness. P60 said, "The structure of the system was the best. It would give an introduction and separate points, then a summary. It's very comprehensive and felt like the best answer to the question. This is what I need to know."

On the other hand, in the No Rhetoric version, which had the lowest rhetorical elements, categories such as informal language, lack of specific details, and logical

structure were identified. These categories were perceived as having many 'false cues' indicative of untruthfulness in the statements. As a result, there was a tendency to judge them as uncertain or false when determining their truthfulness (Table. 21).

Evaluation	Category	Code	Examples
False cues	Insufficient Grounds	no facts, no explanation, vague, insufficient theory, unreasonable, too short, too simple, not technical	"It doesn't provide any clear evidence or proof." "There aren't any facts or explanations given." "The answer from the system is too brief." "The details seem vague." "The AI failed to provide detailed knowledge and information about the topics at hand."
	Informal tone	informal, casual language, tone of voice, not proper academic English, unserious speech style, conversational language,	"It uses informal language and isn't presented as factual." "The way the answer is written makes the information seem unreliable. It's too informal and sounds uncertain." "The language is too colloquial and casual to be credible." "The tone of the text seems unprofessional." "The style is conversational and reminds me of informal conversations that often contain biased answers."
	Unstructured organization	not directly responding, too much side, not answering the question.	"I'm very skeptical about the text because it's so disorganized and unclear." "There are too many exclamations and side comments for it to be taken as factual."
	Human-like	humanistic, like human, distrust, unreliable,	"It didn't feel like the AI was providing facts; it felt almost too human." "It's too human-like to be credible. I wouldn't trust this source to provide the truth." "I feel as though I'm conversing with a human, and for that reason, I find it less trustworthy." "It's like getting an explanation from a casual acquaintance or a laid-back teacher." "It sounds so human yet unreliable, like hearing a friend share anecdote rather than facts. It discourages me from reading the entire response, as the initial sentences already make me skeptical about its reliability." "It's like hearing about something vague that a friend overheard from someone else."

Table. 21. False Cues Used by the Participants to Determine the Falsehood of the AI Response

The overarching sentiment is a lack of trust in the system's outputs, attributed to the casual language, absence of detailed data, and an impression that the AI might be speculating or offering unsubstantiated information. A section of users value straightforward explanations, but others are critical of the colloquial language and the absence of specialized terminology. Users frequently highlight the casual language as unprofessional, which diminishes the perceived credibility. Interestingly, the NR model evoked a feeling of conversing with a friend or teacher, making its outputs seem casual and

potentially unreliable. P57 noted, “It's too personable to be reliable. I wouldn't count on this source for information. It feels like second-hand, vague information from a friend.”

Feedback on Google's responses, which had a mid-level rhetorical component, was mixed. Responses varied in credibility, with some appearing reliable due to detailed and logical explanations and others being seen as vague or unsubstantiated. Descriptions that included dates or events added depth, but some responses seemed too assured or biased, raising accuracy concerns. P3 observed, "Its answers were concise. They are preferable when I'm seeking a quick response."

5.3.2. Expectations for the AI System

Analysis of participants' open-ended responses revealed a consistent discernment criterion: their subjective AI expectations. The alignment between the AI's answers and these expectations significantly influenced their truth judgments. For instance, when evaluating Google's answers, some participants used terms such as "lack of explanation," "insufficient," "do not fit," and "disappointment," expressing dissatisfaction and difficulty in determining the truthfulness of the AI's answers. When discussing the NR version, many participants frequently mentioned that certain parts of the answer did not meet their expectations from the AI. P14 said, "I do not know if it is what I expect from an AI." P5 added, "I don't necessarily think it is false, but it is not an explanation I would seek."

AI responses were often deemed uncertain or false when they did not meet participants' expectations. The NR version's format led to mistrust despite its credible content, whereas ChatGPT's alignment with expectations bolstered trust. Participants

evaluated different AI models using varied criteria: the NR model faced skepticism, especially when individuals lacked prior knowledge, resulting in a dominant false bias. In contrast, ChatGPT secured trust even when conflicting with participants' knowledge, leading to a notable truth bias. Google's answers showed a moderate truth bias, especially when the topic was unfamiliar to participants.

Self-refutation with the NR Response: Participants often perceived the NR responses as false when they did not align with AI expectations. Their judgment began with evaluating content accuracy, linguistic expression, and the expectations for the AI system. Two main perspectives influenced their evaluation. First, dissatisfaction with the linguistic format arose from a casual tone and presentation that did not meet AI expectations. Comments like "words seem unconvincing, affecting credibility" were common. P62 remarked, "Although the response does feel correct, the words being used seem a bit unconvincing, affecting the entire response's credibility." Second, responses that felt 'too humanistic' led to discomfort and distrust. P38 said, "I feel like I am talking with a human, but I cannot trust it that much because of that." Participants often concluded a 'lack of trust' for responses that did not meet their AI expectations.

Self-deception in ChatGPT Responses: Most participants found ChatGPT's answers aligned with their AI expectations, typically deeming them truthful. Their evaluations started with content accuracy, moved to linguistic expression, and concluded with the source's identity. Many appreciated ChatGPT's concise, professional, and well-structured response, boosting its perceived credibility. As P60 stated, "The structured

answer...leaves no room for confusion." There was an overarching belief that AI is precise and factual. This trust was evident when P36 remarked, "AI wouldn't falsify such specifics.". When uncertain about a topic, participants generally trusted the AI, especially if its answer matched their AI expectations.

Notably, the study revealed a fascinating trend: participants often found themselves leaning towards trusting the AI's information, even when it directly contradicted their own knowledge. This intriguing finding points to a significant reliance on machine-generated information, with many decisions stemming from a sense of self-doubt, such as 'I might be mistaken.' P58 encapsulated this, noting, "The answer, with its detailed data, seems certain, even if it contrasts with what I knew. However, other sources could have varying views."

5.3.3. Truth and False Cues in Block 2

In Block 2, participants were tasked with evaluating the truthfulness of AI responses that were filled with a variety of rhetorical elements. The study found that truthfulness was primarily gauged by the answer's structure, detail, and logic. Detailed and coherent answers were generally perceived as more truthful, while those that seemed unsourced or lacking in detail were met with skepticism. Notably, skepticism towards the default version increased when compared to favored versions, like the Footnote version.

Feedback for the Self-correction Version was quite mixed. Some participants felt that the AI correcting itself signaled a level of honesty and reliability. However, for others, these self-corrections raised concerns and introduced elements of doubt. Simple fact modifications by the AI often led to distrust, while broader perspective shifts or

comprehensive clarifications were viewed more positively. P50 added, "The AI system made a correction at the end of the text. While the correction may be true, a reliable AI system shouldn't need to correct itself. Thus, I am skeptical of the provided response."

The Counterargument Version also showed varied reactions. Answers that provided a comprehensive view, detailing various perspectives, were generally well-received. P21 noted, "It gives multiple sides to the argument, giving us a 360-degree approach to the problem." However, some participants believed that these answers were either inconsistent or did not effectively counter opposing views, causing uncertainty about their authenticity.

The Footnote Version was predominantly received positively. Answers presented with footnotes were viewed as more credible and trustworthy. This was attributed to the logical structure, thorough information, and the inclusion of references to reputable sources. Notably, even when some participants identified inconsistencies or minor errors in the footnotes, they still leaned towards trusting the answers, indicating a strong inclination to trust answers with citation references. P59 noted, "While I'm not knowledgeable about the area, I like how the system provided footnotes, making it easy for me to crosscheck the information's truth."

5.3.4. Expectancy for AI System

Participants held diverse expectations for AI, and they tended to use these subjective criteria as a measure for determining truthfulness. The common elements among these expectations are as follows:

Long but concise statement: Participants favored ChatGPT's structured responses, which included intricate details and logical construction, but they generally wished the answers were more condensed. When answers became lengthy, there was a tendency to believe that some content might deviate from the main point.

Balanced Formality: Participants expected AI responses to maintain a formal tone, use professional language, and present ideas in a technical, objective style. Informal or conversational tones were often viewed with skepticism and unease. However, at the same time, they asked the AI to strike a balance, being approachable and easy to understand but not overly casual.

Contextually changing its tone and character: Participants did not always want AI to sound stiff or overly professional. While they expected professional answers for science, history, and sociology questions, they believed a more relaxed tone might be preferred in casual contexts. P51 commented, "The AI should adapt its tone based on the reason someone uses it. Academic topics need formal responses, but more casual subjects can have colloquial answers."

Do not be Human-like and avoid mistakes: Participants felt uncomfortable and disapproving when AI made errors. They believed AI mistakes deviated from the expected precision and meticulousness. These errors made the AI seem human-like and diminished trust. P62 commented, "I have very high expectations for conversational systems. They should use formal language, demonstrate vast knowledge, and avoid mistakes."

Trusted Sources and Transparency: One aspect the participants highly valued in common was the provision of credible sources and footnotes. P52 remarked, "Showing those references convinces me of its validity." The availability of sources is also crucial from a transparency perspective. P56 emphasized, "Providing sources is an excellent form of transparency; it shows where the information originated."

Chapter 6. Discussion

6.1. Findings

This research explored how individuals assessed the truthfulness of AI interactions and investigated the influence of various types of persuasive rhetorical elements on truth discernment. It examined the key points outlined below through quantitative and qualitative methodologies.

6.1.1. *Hypotheses Evaluation*

Since non-verbal and behavioral cues commonly found in human conversation are absent in communication with AI, the study argued that rhetorical elements serve as critical indicators for discerning truthfulness (H2-1). Experimental results aligned with the hypotheses, showing differences in truth discernment based on the presence and intensity of linguistic elements. Individuals were more likely to perceive AI statements as truthful when they contained strong rhetorical elements (H2-2). In Block 1, ChatGPT responses rich in rhetoric elements were perceived as more truthful than Google and NR versions with fewer or no linguistic elements. Interestingly, linguistic elements exerted a more significant influence on truth perception than the actual truthfulness of statements. For instance, false claims from ChatGPT in Block 1 were considered more truthful than accurate claims from Google or NR versions lacking rhetoric elements, indicating the importance of linguistic formalism in people's discernment of truthfulness in statements.

Contrary to the expectation (H1) that discerning truth from falsehood would remain challenging in AI-generated language rich in linguistic elements, individuals exhibited a clear and statistically significant difference between true and false statements when presented (Block 1). In contrast, responses from Google and NR revealed no significant differences in truth discernment between true and false statements. However, there was a bias towards perceiving truthful and false statements as truthful. In Block 2, the footnote version revealed that individuals could discern truth statistically significantly from false responses, and other versions with additional types of rhetoric also tended to perceive truthful responses slightly closer to truth than false responses.

In investigating what rhetorical elements in AI statements could enhance user trust beyond the default responses of AI-generated language models like ChatGPT, which inherently include many rhetorical elements, it was hypothesized that individuals would perceive statements as more truthful when Ethos elements are added (H3). Also, it was expected that people would perceive statements as more truthful when self-correction elements are added (Ethos element introduced) (H3-1). However, contrary to the expectation, individuals tended to perceive statements with such elements as false. Individuals focused on AI making mistakes, using this as a basis for evaluating accuracy with a tendency toward resentment. People apply a rigorous attitude towards mistakes, exhibiting high expectations for AI without making any errors or falsehoods.

It was anticipated that providing sources in footnotes would enhance Ethos and leverage heuristics of professional knowledge, resulting in higher truth discernment (H3-

2), which was confirmed. Footnotes are perceived as trustworthy heuristics, distinctively leading people to judge information as truthful, stemming from the trust individuals place in academic journal references. This trust is grounded in academic journals' rigorous peer-review process, making them challenging to publish easily. Therefore, the information provided with a reference is assumed to be accurate due to the assumption that it has undergone thorough scrutiny (Hilligoss & Rieh, 2008).

Additionally, it was hypothesized that people would perceive statements as more truthful when supplemented with Logos elements(H4). Also, the anticipation was that statements presenting both pros and cons or counterarguments would be perceived as more truthful with elevated Logos element (H4-1). However, compared to ChatGPT's default responses, there was no significant tendency to perceive two-sided statements as more truthful. Nevertheless, two-sided statements were found to be helpful in accurately discerning truth when individuals possessed relevant prior knowledge.

As for the third research question, cognitive judgment tendencies towards language-based generative AI responses (RQ3) were investigated. Consistent with the hypothesis, in Block 1, individuals perceived the system as more trustworthy, logical, and transparent as the richness of rhetorical elements increased (H5-1,2,3). However, contrary to the expectations, as linguistic elements increased, people did not perceive the AI as more human-like or insightful about human thoughts and emotions (H5-4,5). Even if ChatGPT's language adheres to the traditional forms of persuasive rhetoric, people do not perceive it as more human-like or a better understanding of human thoughts and

emotions. Instead, models using casual language with lower persuasive rhetorical elements, whether non-professional or colloquial, were relatively perceived as more human-like.

Finally, the study explored how individual factors, such as prior knowledge, prior experience, and prior trust, interact with message-level rhetorical discernment criteria in detecting deception in AI-generated language (RQ4). Contrary to the expectation that individuals would rely on prior knowledge universally, the influence was model-specific. Contrary to the hypothesis (H6) that individuals with prior knowledge would be able to discern truth based on it accurately, it was revealed that individuals' prior knowledge did not assist in truth discernment from ChatGPT responses. In Block 1, individuals appeared to rely more on the formal, rhetorical elements expressed in the messages provided by the AI than on their prior knowledge for truth discernment. Conversely, in models with relatively fewer rhetorical elements, such as Google and NR versions, individuals tended to perceive familiar topics with their prior knowledge as more truthful regardless of the veracity of the statements. In Block 2, the self-correction version also showed a tendency for individuals familiar with the questions to perceive statements as more truthful, regardless of their actual truthfulness. This pattern was explained as the illusory truth effect, which refers to the heightened probability of perceiving a previously encountered statement as true, a concept linked to the familiarity heuristics discussed in studies on fake news (Pennycook & Rand, 2021).

Regarding prior experience, the hypothesis (H7) that individuals with more experience would have appropriate expectations for the model and, therefore, judge truth more accurately without excessive expectations or disappointments was not supported. The frequency of prior experience did not significantly impact the truth discernment of responses. Looking at the results from Block 1, the frequency of interaction with AI, such as ChatGPT, prior to participation in the experiment did not influence the discernment of ChatGPT responses. However, it was shown to influence biases towards falsehood in NR versions.

Finally, while high prior trust was expected to lead to a strong truth bias towards AI responses (H8), this was not consistently confirmed. In Block 1, prior trust did not influence truth discernment in responses across all models. However, in Block 2, a statistically significant correlation was observed in the footnote version, interpreted as people who generally exhibit high trust in AI also having high trust in traditional heuristic forms of indicators of trust like footnotes, by acknowledging the credibility of scholarly formatted reference (Kim et al., 2019; Luo et al., 2022).

6.1.2. Human Perception of AI

Upon analyzing individuals' perceptions of AI, it was revealed that heuristics-based judgments arise due to the perception of AI as a black box, as discussed earlier (Just & Latzer, 2018; Shin & Park, 2019), along with automation bias (Cummings, 2017; Jones-Jang & Park, 2023; Lee, 2018). Additionally, AI aversion (Alvarado-Valencia &

Barrero, 2014; Bucher, 2017; Dietvorst et al., 2015) was interpreted as limited to specific models within the experiment.

Firstly, people's perception of AI as a black box and judgment based on using heuristics is notable. In previous research, it is noted that due to the lack of knowledge and experience with AI, people tend to perceive AI as a black box and make subjective and limited judgments on AI with their immediate process with heuristics (Just & Latzer, 2018; Shin & Park, 2019). Examining the results of this study, individuals were found to rely on their pre-existing heuristics, developed prior to communication with AI, to discern between truth and falsehood. These heuristics were based on linguistic elements such as structure, logical development, detail, tone, and evidence. Based on these cues, participants tended to rely on Type 1 processing in Dual process Theory, making intuitive judgments through peripheral routes from ELM rather than deliberating on the provided information.

The persuasive rhetorical elements present in highly persuasive ChatGPT responses provided linguistic heuristics that led individuals to perceive them as truthful, even in situations where they contradicted their knowledge. Conversely, the low persuasive rhetorical elements in NR responses provided linguistic heuristics that led individuals to perceive them as false, even when they were consistent with their existing knowledge, thus prompting reliance on intuitive judgments for truth discernment.

Heuristics stemming from Face-to-face conversations were also utilized. In alignment with the findings of previous research (Jakesch et al., 2023), this study also

emphasizes that when interacting with AI, individuals lean on heuristics and biases they have formed from human-to-human communications. P35 explained that "psychologically, a person giving detailed answers seems more informed, leading to a trust in their words". P65 echoed this sentiment, suggesting "it is hard to fabricate detailed information, so richer responses seemed more credible". Even though the generation of detailed content by AI does not necessarily correlate with truthfulness, people tend to perceive detailed responses from AI as truthful. This tendency is reminiscent of the human inclination to consider comprehensive answers as more reliable, a phenomenon supported by Nahari et al. (2014) and Vrij and Fisher (2016). This pattern suggests that our experiences from human conversations directly influence how we judge AI communications. Moreover, the association between truth-telling and providing more detailed, contextual information, as noted by Raskin and Esplin (1991), further illuminates our predisposition to trust detailed AI responses.

Secondly, it was revealed that individuals exhibit automation bias. Individuals perceive AI as a pre-programmed, objective, and consistent system. Additionally, due to the superior performance of AI, there is a belief that content generated by AI is more objective and unbiased than that created by humans, exhibiting automation bias (Cummings, 2017; Jones-Jang & Park, 2023; Lee, 2018). The Experiment findings revealed that when presented with ChatGPT responses, individuals were significantly influenced by their intuitive perceptions of AI's superiority rather than relying on their own knowledge. These decisions indicate automation bias, demonstrating a perception

that AI provides more accurate, objective, and superior knowledge and information than individuals.

Thirdly, AI aversion towards AI appears to persist in a limited manner. Participants' responses indicate that encountering mistakes or errors made by AI in the NR and self-correction versions leads to aversion towards using AI. When participants encountered responses that did not align with their expectations of AI or when they perceived the response as errors or problems in AI, they developed distrust toward AI's outputs, leading them to interpret the responses as biased toward falsehoods. It was interpreted that people exhibit a form of AI aversion (Alvarado-Valencia & Barrero, 2014; Bucher, 2017; Dietvorst et al., 2015) when responding to these versions. However, this aversion towards AI only applies to the specific NR and self-correction versions and did not persist beyond it, so it did not transition into general distrust or aversion towards other versions of the experiments.

Additionally, when examining further the perceptions of users revealed through the experiments, people anticipated AI systems to exhibit computer-like characteristics: clear factual details, technical specifics, and objective presentations. However, when AI demonstrated human-like casual conversational traits, it was met with skepticism. Participants expressed discomfort with AI appearing too human, associating it with potential bias and a lack of objectivity. Many felt that AI should remain neutral, and any hint of human personality or perspective made them doubt its accuracy and truthfulness in responses.

6.1.3. Contextual Influences on Trust and Perception of AI

Participants exhibited a disparity between their professed beliefs and actual trust in ChatGPT. While many voiced skepticisms towards ChatGPT during interviews and expressed strong faith in Google's answers, the data from Block 1 showed the opposite. ChatGPT's responses garnered higher trust ratings than Google's. Even if participants are not explicitly conscious of their AI reliance, their genuine interactions hint at an ingrained trust, perhaps shaped by the rhetorical presentation's perceived credibility. Despite reservations about ChatGPT, the study response suggests a significant trust in this text-based generative AI technology.

In addition, contextual factors emerged as crucial in influencing the discernment of truth in AI statements. The assessment of rhetorical expression proved to be relative depending on the context. In Block 2, the scores for truth discernment for ChatGPT's responses were generally lower than those in Block 1, with a paired t-test revealing $t(264) = 3.34, p = .01$. The observed disparity is likely due to participants interacting with a varied number of AI versions across the two blocks, prompting ongoing comparative assessments. As a result, a trust hierarchy emerged among the versions. ChatGPT, initially the most trusted in Block 1, was surpassed by the Footnote version in Block 2. Thus, trust in these models is relative and dynamic, with context playing a pivotal role in how AI-generated text is judged. Context can significantly affect perceived truthfulness, indicating a need for further investigation into these contextual influences.

6.1.4. Evaluation Process

This study highlights the intricate process individuals utilize in evaluating the truthfulness of AI-generated statements (Figure. 15). Even though people rely on heuristics on rhetoric elements, people do not blindly trust; they make informed decisions considering various aspects and contexts, undergoing an adaptive two-step process. This process can be understood using theories like the ALIED model (Street, 2015) and the Language Expectancy Theory (Burgoon et al., 2002). The ALIED model introduces the concept of an 'individuating cue', which denotes the judgment based on the message format and the alignment of a statement with one's prior knowledge. If this signal provides clarity, truth is determined.

This research findings align with this theory, where individuals in the initial stages examine attitudes, grounds, structure, and completeness of statements to assess their accuracy. If they judge the statement to be sufficiently accurate, they conclude its truthfulness. In this state, linguistic heuristics may be employed to ascertain the accuracy of the statement. Individual factors, such as prior knowledge, common sense, and past experiences, also influence this evaluation. This phase also resonates with Levine's (2014) concepts of 'coherence' (logical consistency of content) and 'correspondence' (content alignment with pre-existing knowledge). P63 stated, "The primary consideration was whether the response made logical sense, especially as some answers were inherently contradictory. I also gauged their alignment with my existing knowledge; familiar information felt more trustworthy than completely unfamiliar information."

If ambiguity persists despite individual cue-based judgments, the phase of judgment based on context generalization is activated. Here, truthfulness is gauged based on how the AI's response meets the user's broader expectations. ALIED refers to this as the 'context general' judgment, rooted in social and cultural contexts that, in digital communications, hinge primarily on linguistic cues. The 'language expectancy model' articulates this idea (Burgoon et al., 2002) deeper, suggesting people's judgments are influenced by language styles that either align with or defy their norms. These linguistic expectations are shaped by broader social, environmental, and contextual factors and participants expect specific linguistic 'dos' and 'don'ts' that influence their perception of an AI's credibility. According to these expectations, people conducted truth judgments of statements; while most participants expressed negative emotions and falsehood biases when encountering casual language styles or errors from AI that did not meet their expectations, a minority of experiment participants expressed leniency or even preference, as falsehood biases were not observed. In conclusion, the veracity of AI statements often seems to be judged progressively based on their logical coherence, alignment with individual factors, and the level of detailed specificity per their expectations.

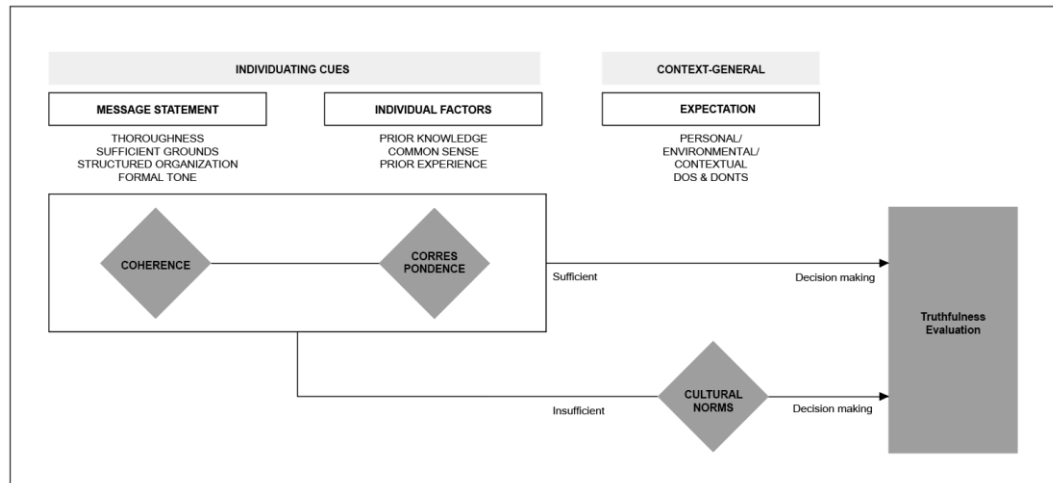


Figure. 15. Adaptive Evaluation Process for AI-Generated Language

6.1.5. Contribution

This study explored the impact of linguistic forms in text-based AI design on users' ability to discern truth, revealing a close relationship between trust in AI responses and rhetorical elements. It was found that AI outcomes, crafted through reinforcement learning with human feedback, incorporate elements of traditional rhetorical argumentation. This usage of rhetoric elevates the perceived trustworthiness of AI-generated language, exposing users to more sophisticated and covert manipulations and deceptions. This study's contributions to the advancement of truth discernment in AI communication and the understanding of deception in text-based AI systems are threefold. First, the research suggests viewing and addressing problems in text based generative AI from a rhetorical perspective for transparent and truthful human-AI communication. Second, it sheds light on how various rhetorical techniques influence user judgment and highlights the need for subsequent linguistic research, guidelines, and

regulations on language usage to foster a proper mutual trust relationship with users. Finally, the study provided insights into the cognitive heuristics, biases, and decision processes humans employ when assessing AI language. This research also offers direction for text-based AI design in the industry by clarifying what people expect from AI.

6.2. Limitations

While this study has many remarkable implications, it also has limitations. First, it primarily focuses on Aristotle's rhetorical elements of 'Logos' and 'Ethos' but omits 'Pathos', which pertains to emotional appeal. Therefore, the findings should be contextualized within these chosen elements.

Second, this study utilized a text-based AI simulator, focusing on the presence or absence of rhetorical elements. When these elements were removed using ChatGPT, the language shifted to a more colloquial tone. Participants noted discomfort with this colloquialism in a truth-judgment setting. There's a need to investigate how outcomes might differ if conducted via real-time AI speakers rather than text.

Third, while LLM studies encompass various research topics like text classification, text generation, inference, data extraction/transformation/enhancement/processing, code debugging, generation evaluation, and machine translation (Liu et al., 2023; Lu et al., 2023; Peng et al., 2023), this study specifically conducted experiments within the context of question and answering using the ChatGPT Language Model. As a

result, it may be challenging to generalize the findings of this study to all Language Models and different contexts. However, upon examination in comparative studies with models such as ChatGPT and BERT, it was observed that there are context-specific performance differences among the models. Nevertheless, in the aspect of question and answer, which is the focus of this study, the models exhibited similar performance (Zhong et al., 2023). Therefore, even though this research was conducted with a limited LLM model (ChatGPT), it cannot be solely defined as a model-specific characteristic. Hence, this study holds significance as a starting point, and expanding the scope of research through various case studies applicable to general contexts and diverse LLM models will be necessary in the future.

In addition, further research may be needed to investigate linguistic factors that may influence truth discernment, such as message length or other language-related elements. While previous research on LBC explored the role of detailed linguistic factors such as length, complexity, diversity, and uncertainty as cues, the results were inconsistent (Rubin & Conroy, 2012). Moreover, the application of these cues to the sentence structure and persuasive rhetoric of ChatGPT, which is the focus of this study, may not be applicable. To mitigate the potential influences of these factors on the truth discernment outcomes of this experiment, all feasible variables were controlled for. The only variable beyond control was the length of responses. In the results of Block 1, the comparatively longer responses from ChatGPT exhibited significantly higher trustworthiness compared to the shorter responses from Google and the No Rhetoric

version. However, in the Block 2 experiment, all responses were based on ChatGPT, resulting in uniformly longer responses. Therefore, it cannot be conclusively asserted that the length of responses is a determining factor in truth discernment for ChatGPT language, as meaningful differences in truth discernment occurred due to variations in rhetorical elements included in the responses, despite the uniform length observed in the Block 2 experiment.

6.3. Suggestion for Future Work

This study holds significance in exploring the impact of linguistic forms in text-based AI design on users' ability to discern truth, revealing a close relationship between trust in AI responses and rhetorical elements. However, through various subsequent studies, further academic and industrial contributions can be expected. These studies include the following.

The first follow-up research topic could aim to examine the changes in users' perceptions and truth discernment abilities when time and continuity are involved in communication with AI. This study was designed in a question-and-answer format for single-turn interactions, which may differ from multi-turn experiences typically encountered by users in everyday LLM interactions. In multi-turn interactions, users engage in ongoing dialogue, have opportunities to provide feedback to the system, and interact with AI continuously. Users can ask successive follow-up questions, receive updated responses from AI, point out errors in the information provided, or engage in debates, thus assessing various aspects of communication abilities with ChatGPT,

including its information delivery, contextual understanding, interpersonal conversational skills, and personalization capabilities. Through these processes, users can develop a sustained cognitive trust relationship rather than temporary cognitive perception established through single-turn interactions.

For example, in the context of information delivery and truth discernment perspectives, allowing users to conduct 4-5 multi-turn interactions where they can inquire about detailed aspects of received information may lead users to generate different perspectives on ChatGPT's responses. One such example could involve evaluating rhetorical elements such as self-correction. In this experiment, users deemed responses containing self-correction inappropriate in the context of single-turn interactions. Many users felt uncomfortable with the fact that AI produced errors, while others found it difficult to accept the fact that within a single response, the information provided earlier was modified after an error was made and corrected with a slight time delay. Due to ChatGPT's characteristic of providing responses line-by-line in real-time, the fact that AI self-corrects without user intervention within the same turn seemed awkward. Experiment participants questioned why AI couldn't provide the correct answer from the beginning if it was going to be corrected later.

However, if such correction processes are clearly provided within a multi-turn interaction with a time gap during turn-taking, similar to the recognition of errors and confession later in human conversations, it may yield different outcomes than the single turn case. Furthermore, continuity over several days, rather than a one-time interaction,

allows for the creation of conversation context, the development of ChatGPT's character through participation in dialogue, and the potential formation of a social relationship between individuals and ChatGPT. Through changes over time and continuous interactions, important insights into how emotional trust evolves while sharing personal thoughts and feelings can be identified.

The second follow-up research topic necessitates investigation into how AI responses are managed within constrained form factors like mobile devices. Unlike search engines, large language models (LLMs) typically provide one relatively lengthy response using narrative and descriptive rhetorical techniques, relying on linguistic elements rather than offering a variety of search result options with links. Interpreting such textual content requires significant temporal and perceptual resources due to the need to process detailed information and narratives during the provision of responses line by line, including the waiting time during this interaction. This form of interaction is considered unsuitable for form factors characterized by temporal immediacy and spatial limitations, such as mobile screens. To ensure that users can understand the messages created by LLMs within a limited form-factor screen, specific research is required to determine the appropriate amount of information and methods for delivery and the provision of interfaces. This research should address how to effectively decide and convey the right amount of information for optimal user comprehension within the context of mobile and other constrained form factors.

The third area of research entails investigating how perceptions of AI, including deception detection and trustworthiness, will evolve in modalities beyond text-based

communication, such as spoken interactions. In spoken language interactions, various capabilities emerge, including detecting and analyzing gender, age, and emotion based on individuals' voice tone, speed, and intonation when interacting with LLMs. Similarly, LLMs themselves can adopt specific voices and non-verbal cues like gender, tone, and speed to enhance interaction with humans. This process may also involve establishing consistent characters to improve coherence and provide flexible communication and trustworthy interactions across diverse modalities. Moreover, applying named embodied agents in these processes introduces elements such as human likeness, intimacy, and familiarity, necessitating further research into how these factors influence people's perceptions and truth discernment of AI.

The fourth follow-up research topic aims to measure the changes in AI responses and the corresponding shifts in user perception and discernment based on different prompts. While this study provided responses from default ChatGPT based on a predetermined style of questions, it's essential to note that the quality and context of responses can vary significantly based on how prompts are engineered.

A prompt is the text input provided to guide output generation by the Large Language Model (LLM). The characteristics of the prompt can significantly influence the quality of the response generated by LLM. LLM adjusts its responses based on subtle patterns in the prompt, such as instructions and examples, thereby being influenced by the wording, selection, and order of words in the prompt. The objective of prompt engineering

is to enhance the performance, accuracy, and usability of LLM by creating appropriate prompts tailored to produce the desired results for users.

In Industrial landscape there are Prompt Engineering methods like Chain of Thought (breaking down questions into smaller logical components), Tree of Thought (generating and executing possible next steps), Complexity-Based (selecting rollouts with the longest chain of thought), Generated Knowledge (generating necessary facts first), and Self-Refine (iterating problem-solving and criticism processes) (Fu et al., 2022; Madaan et al., 2024; Shi et al., 2020, Wei et al., 2022; Yao et al., 2024).

AI may produce different outputs for the same topic based on various linguistic styles, questioning methods, and contextual settings that individuals may unconsciously employ. It would be interesting for future research to explore how AI outputs are generated based on users' diverse rhetorical prompting elements and how the characteristics of such outputs affect users' truth discernment and other types of cognitive perception and decision-making processes.

Chapter 7. Conclusion

Since the release of user-customizable GPTs, it has become possible for anyone to create personalized AI that meets a wide array of needs in daily life. Customizable AI can be created through simple instructions without coding or complex processes. These custom GPTs can be shared in the GPT store, where users can also explore and use GPTs created by others. The GPTs and Store have unlocked new AI possibilities that can be professionally utilized in various aspects of life by adding creativity to the expertise of each user.

Additionally, with the ability to process multimodal data in forms such as text, voice, and images, it is now possible to offer more comprehensive data analysis and intuitive interfaces, providing a natural and enhanced user experience. Multimodal interactions in fields like education and healthcare can improve effective information delivery and increase the success rate of diagnosis and treatment. In the realms of design and entertainment, they hold the potential to generate creative solutions.

However, as AI becomes increasingly applicable and effective across diverse life domains, a more cautious approach towards AI is necessary. It is crucial to recognize and address the potential risks, such as biases in the various data sources used for AI development that might lead to incorrect predictions or decisions, the misuse of personal information contained in image and voice data, and the issues related to the truthfulness and accuracy of the provided information. Understanding and mitigating these risks is

essential for realizing the full potential of AI while safeguarding against its inherent dangers.

With the advent of AI, we are bound to encounter various levels of misinformation and deception in our daily lives, whether intentional or unintentional. According to the results of this experiment aimed at measuring users' ability to discern the truthfulness of AI-generated content, AI responses that are logically structured and include detailed investigative elements are perceived as truthful. This implies that as we increasingly engage with various AI-integrated services in the future, we may be exposed to convincing falsehoods, posing a constant challenge in distinguishing the truth. In the context of truth discernment in AI language, participants in these experiments relied on heuristics built from their human interactions. Recognizing that our judgment abilities are susceptible becomes a crucial concern, given the unconscious connection between immediate and intuitive judgments related to truth discernment and the subconscious trust associated with rhetoric formats.

As AI language becomes more akin to human natural language, the realm of unconscious judgments relying on linguistic heuristics may no longer be trustworthy and safe. We remain unaware of how the heuristics for our current language might be applied to AI language and what risks that may pose. Presently, the hallucination and misinformation in AI result from systemic limitations; however, in the future, intentional disinformation may manipulate user safety and values through the reverse application of linguistic heuristics. Nevertheless, the fundamental safety net for users, ensuring that the

linguistic form of AI output is directed in a non-harmful direction for humans, is not in place.

Therefore, individuals must develop digital literacy skills, enabling them to critically perceive, effectively utilize, and evaluate the information they encounter (Garner, 2006). It is essential to recognize signals of accurate truth discernment, exercise judgment, and cultivate reflective and contemplative thinking. Moreover, to cope with the continuous emergence of new formats of information and deception in the digital world, information literacy skills need to be continuously updated (Karlova & Fisher, 2013). Additionally, education is required on two fronts: first, raising awareness of the potential for LLMs to generate falsehoods and educating individuals to prevent the dissemination of such false information; second, educating individuals on the characteristics of AI-generated language that may lead them to misconstrue falsehoods as truths, and addressing human cognitive errors that enable this.

AI researchers need to clearly understand the cognitive risks associated with using rhetorical expressions through precise studies that assess the intensity and scope of these risks. Systematic research is required to determine which types of rhetoric instill the most trust, regardless of the truthfulness of the statements, and to identify elements with deceptive properties. Understanding how rhetorical expressions impact human cognition and how linguistic heuristics manifest and operate in interactions with AI is crucial.

AI designers must strive to create a safe user environment through appropriate interventions from the user's perspective. Based on research, it is essential to issue

appropriate warnings about cognitive vulnerabilities, indicate sources and accuracy, and implement proactive measures to prevent unconditional and unconscious trust. It is also imperative for AI developers to establish and adhere to guidelines and regulations governing the application of rhetoric throughout the development process. Utilizing human feedback that meets these standards in processes such as prompt generation for large language models and output labeling is vital. Additionally, compliance with these guidelines in data selection, reward model construction, and training processes is necessary.

The AI industry must recognize the heuristic biases and trust that people have in rhetorical expressions and bear an ethical and moral responsibility to clearly inform users of potential risks instead of exploiting these biases. During research and development, it is crucial to implement systems to ensure fair use of data and prevent biases. AI content should be filtered during the production and distribution of services, and the spread of harmful content should be prevented. From a user management perspective, it is necessary to provide education on how AI systems operate, their functions, biases, and issues and offer control measures such as privacy settings, transparency features, and content filtering.

In conclusion, this study emphasizes the importance of linguistic research to enhance transparency, promote truth discernment in human-AI interactions, and provide actionable strategies for user protection and a safe environment. By providing actionable strategies for user protection and fostering a safe environment, we can mitigate the

cognitive risks associated with AI-generated content. Furthermore, this research outlines a comprehensive roadmap for the AI field, emphasizing the need for ethical practices, rigorous standards, and continuous education to ensure that AI technologies develop in a manner that respects and safeguards human cognitive processes. By addressing these challenges, we can build a future where AI serves as a trustworthy and reliable tool, enhancing rather than undermining human judgment and understanding.

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국문 초록

Deciphering Deception: How Different Rhetoric of AI Language Impacts Users' Sense of Truth in LLMs

AI 언어에 사용된 수사적 표현이 사용자들의 진실 판별에 미치는 영향에 대한 연구

유 다 혜

UX 전공

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사용자들은 가상 비서(virtual assistants), 자동 완성(autocomplete), 스마트 답변(smart replies), 뿐만 아니라 ChatGPT와 같은 생성형 AI의 발전으로 일상에서 더욱 다양한 AI 생성 언어를 경험하고 있습니다. 이는 사용자들이 새로운 형태의 기만, 의사소통 위험, 조작에 노출되어 있다는 것을 의미합니다. 사용자들은 ChatGPT와 같은 대화형 AI 시스템이 만들어낸 문장의 진위 여부를 판별하는 데 어려움을 겪고 있습니다. 본 연구는 사용자들이 AI 생성 언어의 진위를 판별하기 어려워하는 이유를 수사학적 관점에서 조사하였습니다. 세 가지 레벨의 수사학적 요소가 반영된 AI 모델 간 답변의 진위 판별도 비교 실험(Block 1), 네 가지 다른 설득적 수사학 요소를 반영한 AI 모델 간 답변의 진위 판별도 비교 실험(Block 2), 그리고 인터뷰를 통해 AI 시스템별 사용자의 진위 판별 능력 차이 및 각 판별에 사용된 근거에 대한 조사가 이루어졌습니다. 사용자들은 수사학적 요소가 적을수록 진실과 거짓을 유의미하게 구분하지 못했고, 수사학적 요소가 많을수록 강한 진실 편향을 보이며 거짓된 진술도 진실로 판단했습니다. 또한 사용자가 갖고 있는 AI 시스템에 대한 기대치도 진실

판별에 영향을 미쳤는데, 기대치에 부합할 때는 AI의 답변이 자신의 의견과 달라도 신뢰하며 진실 편향을 보였고, 기대치에 부합하지 않을 때는 신뢰하지 않는 거짓 편향을 보였습니다. 또한 구어체, 캐주얼한 단어가 사용되어 사람과의 대화를 연상시키는 경우는 진술이 거짓이라고 인식하는 경향이 있었습니다. 다양한 수사학적 상세 요소 중에서는 주석이 제공될 경우, 거짓 답변도 진실로 인식하는 높은 진실 편향을 보였고, 반론이 제공될 경우 사용자의 사전 지식이 유용하게 사용되어 보다 정확하게 진위를 판단할 수 있었습니다. 사람들은 AI와의 소통 시 심사숙고하는 대신 수사학적인 휴리스틱에 기반하여 직관적으로 판단을 내렸습니다. 사람들은 AI를 객관적이고 우월하다고 인식하며, 자신의 지식보다 더 선호하는 경향을 보였으나 AI 응답에서 오류를 발견할 경우는 AI에 대한 부정적인 인식과 거짓 편향을 보였습니다. 본 연구 결과는 AI 생성 언어의 진위 여부와 관계없이 사람들이 수사학적 요소에 따라 답변의 진위를 판별하는 데 매우 취약하거나 편견을 가지고 있다는 것을 보여주었습니다. 또한 사람들이 더 진실로 믿을 수 있도록 AI 시스템을 인위적으로 개발할 수 있는 조작 가능성을 보여줍니다. 사용자를 기만하는 속성을 제거한 투명한 Human-AI 커뮤니케이션을 위해서는 대화형 인공지능 시스템 디자인 시 사용자의 진위 판별 능력을 향상시키는 수사학적 요소의 발굴과 적절한 사용, 그리고 사용자 기만성이 높은 답변 유형에 대한 가이드라인을 제공하는 등 여러 대안적 모색이 필요합니다. 사용자들은 AI가 거짓을 생산할 수 있다는 인식을 가져야 하며, 사람이 가지고 있는 언어적 휴리스틱과 인지적 편향에 대한 이해하는 것이 필요합니다. AI 생성 언어에서 정확한 진실 판별의 신호를 인식하고, 정보를 비판적으로 평가할 수 있는 디지털 소양을 갖추는 것이 중요합니다.

키워드: AI 생성 언어; 인간-AI 커뮤니케이션; 속임수 탐지; 수사학; AI 인식; 휴리스틱